

Dual-Mode LED Aided Visible Light Positioning System Under Multi-Path Propagation: Design and Demonstration

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Abstract—In this paper, we propose a novel visible light positioning (VLP) scheme under multi-path propagation, which is a practical scenario that has not been well studied for VLP. The new scheme exploits the so-called dual-mode light-emitting diode (DM-LED) of different radiation lobe mode numbers at the transmitter and a photodiode at the receiver. Specifically, we devise a method that utilizes a radiation angle measurement approach for DM-LED, derive the Cramér-Rao lower bound (CRLB) of the estimated distance and analyze the characteristics of key parameters. In addition, two localization algorithms based on linear and nonlinear least squares methods are developed. Simulation results show that under the ideal light-of-sight scenario, the CRLB of the proposed VLP scheme can be close to that of the conventional received signal strength (RSS) aided VLP scheme. Furthermore, we implement the first ever prototype DM-LED lamp to validate the new DM-LED aided VLP system model. Both simulation and experiment results demonstrate that the proposed system outperforms its RSS-aided VLP counterpart in terms of positioning accuracy in the more realistic, and thus more challenging multi-path scenario, even with a tilting receiver.

Index Terms—Visible light positioning (VLP), dual-mode LED, multi-path propagation, received signal strength ratio (RSSR), radiation angle measurement, radiation lobe mode number (RLMN), Cramér-Rao lower bound (CRLB).

I. INTRODUCTION

IN RECENT years, visible light communication (VLC) based on light emitting diode (LED) has attracted great attention and emerged as a promising wireless technology [1]. Supporting simultaneous illumination and communication, VLC can be readily exploited as a viable solution for indoor location based services [2]. Compared with the conventional indoor positioning schemes based on radio-frequency technologies, visible light positioning (VLP) has many advantages including low deployment cost, easily controllable interference, free from electromagnetic pollution, etc., all rendering its superior applicability to the indoor location based services scenario. Furthermore, the VLP technologies employing

photodiode (PD) as the receiver front-end, namely the so-called non-imaging VLP (nVLP) schemes [1], [2], are among the most promising candidate solutions for practical VLP applications. Popular nVLP schemes include those based on the received signal strength (RSS) [3]–[5], time-of-arrival [6], time-difference-of-arrival [7], [8], angle-of-arrival [9], [10] and other hybrid mechanisms [11], [12].

However, in practical scenarios, due to the ubiquitous light reflections in indoor environments, the impact of multi-path propagation effect on the achievable performance of nVLP systems may not be negligible under many circumstances. In the literature, there have been some investigations on this issue. For example, some researchers show that multi-path reflections have a large impact on RSS-based nVLP systems [13]–[17], where the positioning performance deteriorates quickly when the user equipment (UE) gets close to the corners of the room. In addition, an experiment in a warehouse demonstrates that the multi-path reflections caused by the storage shelf rack can degrade the positioning accuracy [18]. Furthermore, in [19], it is theoretically confirmed that the positioning performances will hit an error floor caused by the unknown non-line-of-sight (NLOS) links. However, these investigations only analyze the effect of multi-path reflections on positioning accuracy, though without proposing a measure to mitigate such an effect. Aiming to tackle this issue, some fingerprint-based methods [20]–[22] have been proposed. Nonetheless, they require pre-measurement of RSSs as fingerprints or training data. When the environment changes, new measurements have to be done, which is costly from practical perspectives. Compared to fingerprint-based solutions, the non-fingerprint methods are usually more cost effective. Among the very few non-fingerprint solutions considering the adverse effect of multi-path reflections on positioning accuracy, the authors of [23] suggest to exploit the top few signals with the highest RSSs to calculate the UE's position. Nevertheless, the NLOS components remaining in the selected signals still have a non-ignorable impact on the system performance. Later, a line-of-sight (LOS) and NLOS identification algorithm is considered in [24]. However, the method is based on time division multiplexing, which imposes a stringent requirement on inter-LED synchronization. Based on the above background, we can see that the existing work on mitigating multi-path reflections in nVLP applications is still in the preliminary stage.

A complete solution to a large-scale VLP problem, for example invoking a positioning service in a large area deployed with many positioning lamps, includes the detection

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of the LED-IDs and the calculation of the UE's spatial coordinates. In an nVLP system, the LED-IDs can be obtained by some existing methods [25]–[29]. Thus, in this paper we restrict our focus on how to effectively resolve the UE positioning problem in nVLP systems, particularly under the multi-path propagation scenarios. Compared with conventional RSS-based nVLP (RSS-nVLP) schemes employing single-mode LEDs, we propose to exploit the so-called dual-mode LEDs (DM-LED) with different radiation lobe mode numbers (RLMNs), resulting in the new dual-mode nVLP (DM-nVLP) system, which was first introduced in our preliminary work in [30]. Our new work compared with [30] includes the robust frequency division multiplexing based mechanism, a comprehensive theoretical analysis on reflection and DM-LED models, the derivation of the Cramér-Rao lower bound (CRLB), extensive simulation and experiment results, etc. *To our best knowledge, we are the first to propose the DM-LED model, implement the first ever prototype DM-LED lamp and demonstrate the new DM-nVLP system. With the aid of the comprehensive yet efficient non-fingerprint solution, the impact of multi-path propagation effect can be effectively mitigated, significantly improving the positioning performance up to an average accuracy of about 4 cm and 6 cm in simulations and experiments, respectively.* Our main contributions are:

- We propose the novel DM-nVLP concept in the context of practical multi-path propagation scenarios. Based on an in-depth analysis of the DM-LED model, we devise a radiation angle measurement mechanism exploiting the so-called received signal strength ratio (RSSR), which can provide a reliable estimate of the radiation angle between the DM-LED lamp and the UE.
- We derive the CRLB of the accuracy of the estimated distance for the DM-nVLP model. It offers an important guideline for selecting key system parameters, such as the optical power of the DM-LED, the number of required measurements and the inter-lamp distance.
- We provide a novel numerical approach to analyze the main parameters related to multi-path propagations in the DM-nVLP model. Based on the iterative site-based model [31], [32], we further evaluate our approach under scenarios with low to high reflections, where key characteristics of the system are identified and then exploited in the positioning algorithm design.
- We devise two positioning algorithms that can cope with multi-path reflections, namely the DM-LED aided nonlinear least squares (DM-NLS) and the DM-LED aided linear least squares (DM-LLS) algorithms, which solve the nonlinear and linear constrained minimization problems regarding the LED radiation angle, respectively.
- We implement the first prototype DM-LED lamp supporting multiple RLMNs, noting that no DM-LED product is available in the market. We then develop the first proof-of-concept DM-nVLP system and validate it with extensive simulations and experiments.

The rest of the paper is organized as follows. In Section II, we introduce the proposed DM-nVLP model taking into account the multi-path propagation effect, and discuss the radiation angle measurement framework together with details

of some key parameters. Then in Section III, two positioning algorithms, namely the DM-NLS and the DM-LLS schemes are introduced, followed by the CRLB analysis of the estimated distance between the LED lamp and the UE in Section IV. The simulation results of the proposed schemes are provided in Section V, while we demonstrate a prototype DM-nVLP system and provide abundant experiment results in Section V-B. Finally, our conclusions are outlined in Section VI.

II. THE PROPOSED SYSTEM MODEL

A. Model Formation

Assume that the i -th ($i = 0, \dots, V - 1$) lamp is equipped with J LEDs of different RLMNs, and that each LED conforms the Lambertian radiation pattern with the uniaxial symmetry property. Then the generalized LED radiation intensity from the τ -th LED ($\tau = 0, \dots, J - 1$) of the i -th lamp, denoted by $I(\theta_i, n_\tau)$, is given by [33]

$$I(\theta_i, n_\tau) = \frac{n_\tau + 1}{2\pi} \cos^{n_\tau}(\theta_i), \quad (1)$$

where θ_i is the i -th lamp's radiation angle, n_τ ($\tau = 0, \dots, J - 1$) is the RLMN representing the directionality of the light source. For simplicity, we assume each of all lamps employs the same set of J LEDs with the same set of RLMNs. Thus, we ignore the lamp index i in n_τ for notation brevity. Then, we have [33]

$$n_\tau = -\frac{\ln 2}{\ln(\cos \theta_{\frac{1}{2}, \tau})}, \quad (2)$$

where $\theta_{\frac{1}{2}, \tau}$ is the half-power semi-angle of the LED with RLMN n_τ . Note that the RLMN n_τ is related to $\theta_{\frac{1}{2}, \tau}$. For example, a Lambertian transmitter with $\theta_{\frac{1}{2}, \tau} = 60^\circ$ and a typical directional transmitter with $\theta_{\frac{1}{2}, \tau} = 15^\circ$ correspond to $n_\tau = 1$ and $n_\tau = 20$ [33], respectively.

As an example of the proposed DM-nVLP model, we assume that each of the DM-LED lamps, serving as positioning anchors, is equipped with two LEDs with different RLMNs, $n_\tau \in \mathbb{R}^+$, $\tau = \{0, 1\}$, $n_0 \neq n_1$. Without loss of generality, we assume $n_0 < n_1$ in the sequel. In each DM-LED lamp, both LEDs are installed adjacently on the lamp's luminous surface, as illustrated in Fig. 1. The mid-point between the two LEDs is used to represent the location of the DM-LED lamp. The frequency division multiplexing mechanism is adopted to separate the received optical signals transmitted from multiple LEDs. More specifically, each LED of each DM-LED lamp is modulated by an on-off keying signal of a unique frequency at a peak optical power of P_L .

Without loss of generality, we now focus on the i -th DM-LED lamp, whose center is assumed to be at a location $S_i = (x_i, y_i, z_i) = (x_i, y_i, h_R) \in \mathbb{R}^3$, where x_i , y_i and z_i refer to the labels of the three-dimensional coordinate system for the indoor room model, while h_R represents the vertical distance between the ceiling and the floor. Additionally, a UE equipped with a PD is placed in the room, where we assume the PD's center is located at $U = (x_U, y_U, h_U) \in \mathbb{R}^3$, with h_U being the vertical distance from the PD to the floor. The distance between the lamp and the PD is denoted by $d_i =$

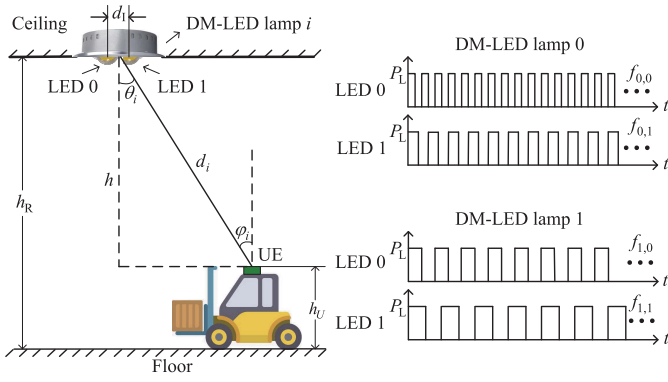


Fig. 1. The proposed DM-nVLP system model.

$\|S_i - U\|_2$. As a typical case, we assume that the DM-LED lamps point downwards and the PD points upwards. Note that if a PD tilt is introduced, our scheme remains the same except that the tilting angle, as an offset, needs to be considered in the coordinate system when calculating for example φ_i , where φ_i is the incident angle between the i -th lamp and the PD. Based on the geometric topology in Fig. 1, there exists $\theta_i = \varphi_i$, and

$$\cos \theta_i = \cos \varphi_i = \frac{h}{d_i}, \quad (3)$$

where $h = h_R - h_U$. It is assumed that h remains unchanged when the UE moves, which can be easily fulfilled by attaching the PD to those UEs, such as navigation trolleys, shopping carts, factory forklifts and industrial robots, for corresponding VLP application scenarios. Note that given h and θ_i , it is easy to derive d_i , the key parameter for UE positioning. Furthermore, let d_1 be the distance between the two LEDs of the same DM-LED lamp, which is usually up to 1-2 cm, about 0.5-1% of a typical d_i . For simplicity, we assume $d_1 \approx 0$ in the derivations of the theoretical DM-LED system model, which will be introduced later. However, a value of $d_1 = 2$ cm is used in our simulations and experiments in Section V.

B. Multi-Path Propagation

In practical indoor scenarios, the light rays emitted from the LED lamps are subjected to various reflections due to the existence of indoor facilities, for example walls, ceiling, floor, furniture, and even human bodies. According to [31], [32], the total light power from multi-path propagation is comparable with that from LOS propagation. Furthermore, in a highly-reflective indoor environment, the contribution to the total light power from multi-path propagation may be even greater than that from LOS transmission. Objects with different reflective surfaces typically have different reflection coefficients. The most significant reflections often come from for example the areas close to walls, such as the room corners. Naturally, the signal strength measured by the VLP receiver is a superposition of the transmitted signal's multi-path propagations, and depends on the room's specific geometry as well as the distribution of the objects inside the room.

Considering a typical rectangular room model, the factors affecting reflection strengths may be defined as an environmental parameter set as

$$E_N = \{l_R, w_R, h_R, h_U, \varrho_W, \varrho_E, \varrho_S, \varrho_N, \varrho_C, \varrho_F\}, \quad (4)$$

where $l_R, w_R, h_R, \varrho_W, \varrho_E, \varrho_S, \varrho_N, \varrho_C$ and ϱ_F denote the length, width, height, west/east/south/north wall, and ceiling/floor reflection coefficients of the room, respectively. Considering multi-path propagations, the channel gain between the i -th lamp and the PD can be written as

$$\begin{aligned} H(S_i, U, E_N, n_\tau) &= \sum_{k=0}^{\infty} H^{(k)}(S_i, U, E_N, n_\tau) \\ &= H_{\text{LOS}}(S_i, U, n_\tau) + H_{\text{NLOS}}(S_i, U, E_N, n_\tau) \\ &= H^{(0)}(S_i, U, n_\tau) + \sum_{k=1}^{\infty} H^{(k)}(S_i, U, E_N, n_\tau) \\ &\approx H^{(0)}(S_i, U, n_\tau) + \sum_{k=1}^K H^{(k)}(S_i, U, E_N, n_\tau), \end{aligned} \quad (5)$$

where $H^{(k)}(\cdot)$ represents the channel gain between the lamp and the PD when the light undergoes k reflections, with $k = 0$ indicating the LOS path, while $H_{\text{LOS}}(\cdot)$ and $H_{\text{NLOS}}(\cdot)$ refer to the LOS and the NLOS channel gains, respectively. Since the contributions to the overall channel gain by high-order reflections are typically small, the NLOS part of (5) is approximated by up to a small-to-medium value of K reflections only [34]. Moreover, note that the LOS channel gain $H_{\text{LOS}}(S_i, U, n_\tau) = H^{(0)}(S_i, U, n_\tau)$ does not depend on E_N . According to the Lambertian radiation property, it can be formulated as [32]

$$H_{\text{LOS}}(S_i, U, n_\tau) = I(\theta_i, n_\tau) \cos^m \varphi_i \frac{A_U}{d_i^2} \mathbf{F}(\varphi_i), \quad (6)$$

where $I(\theta_i, n_\tau)$ is defined by (1), A_U is the area of the PD, and $\mathbf{F}(\varphi_i)$ represents an indicator function which takes one for $\varphi_i \leq \varphi_{\text{FOV}}$ and zero otherwise, with φ_{FOV} being the field of view (FOV). Furthermore, we have $m = -\ln 2 / \ln(\cos \varphi_{\frac{1}{2}})$ [35], where $\varphi_{\frac{1}{2}}$ is the half-power semi-angle of PD. For notation simplicity, we use $\mathbf{F}_i$ to replace $\mathbf{F}(\varphi_i)$ in the sequel.

On the other hand, since the channel gain of an NLOS link, H_{NLOS} , is related to the nondeterministic function E_N and the specific combination of $\{S_i, U\}$, it is difficult to obtain its closed-form expression. Nonetheless, for each fixed location in the room, if the surrounding environment does not change, the reflective paths may be considered as static. In this case, H_{NLOS} may be approximated as a positive constant subjected to $\{S_i, U, E_N, n_\tau\}$. Inspired by the multiplicative uncertainty model [36, (53)], we introduce an artificial parameter called the channel gain ratio $\rho(S_i, U, E_N, n_\tau)$. Specifically, we define the NLOS channel gain as

$$H_{\text{NLOS}}(S_i, U, E_N, n_\tau) = \rho(S_i, U, E_N, n_\tau) H_{\text{LOS}}(S_i, U, n_\tau). \quad (7)$$

The channel gain ratio defined in (7) provides a way to link the LOS and the NLOS channel gains, creating a connection

helpful for deriving the theoretical model of the DM-nVLP system later.

C. Measurement of Radiation Angle

Based on the Lambertian radiation property [32], the system model of Fig. 1, and equations (1), (3), (6) and (7), the received peak signal emitted from the τ -th LED of the i -th DM-LED lamp during the q -th ($q = 0, \dots, Q-1$) measurement may be expressed by

$$\begin{aligned} u_{i,q}^{(\tau)} &= \beta \ell P_L [H_{\text{LOS}}(S_i, U, n_\tau) + H_{\text{NLOS}}(S_i, U, E_N, n_\tau)] + z_q^{(\tau)} \\ &= \beta \ell P_L [1 + \rho(S_i, U, E_N, n_\tau)] H_{\text{LOS}}(S_i, U, n_\tau) + z_q^{(\tau)} \\ &= \beta \ell P_L I(\theta_i, n_\tau) [1 + \rho(S_i, U, E_N, n_\tau)] \cos^m \varphi_i \frac{A_U}{d_i^2} \mathbf{F}_i + z_q^{(\tau)} \\ &= \beta \ell P_L \frac{n_\tau + 1}{2\pi} \cos^{n_\tau} \theta_i [1 + \rho(S_i, U, E_N, n_\tau)] \\ &\quad \times \cos^m \varphi_i \frac{A_U}{d_i^2} \mathbf{F}_i + z_q^{(\tau)} \\ &= \beta \ell P_L \frac{n_\tau + 1}{2\pi} \cos^{(n_\tau+m+2)} \theta_i \\ &\quad \times [1 + \rho(S_i, U, E_N, n_\tau)] \frac{A_U}{h^2} \mathbf{F}_i + z_q^{(\tau)}, \end{aligned} \quad (8)$$

where β is the PD responsivity, ℓ is the amplifying coefficient, $\tau = \{0, 1\}$, and $z_q^{(\tau)}$ is modeled as an additive white Gaussian noise (AWGN) random variable with zero mean and variance σ^2 [3].

Furthermore, for the convenience of the derivations in Section IV, we transform (8) to vector form as

$$\mathbf{u}_{i,q} = \mathbf{k} \mathbf{c}_i + \mathbf{z}_q, \quad i = 0, \dots, V-1; \quad q = 0, \dots, Q-1, \quad (9)$$

where $\mathbf{u}_{i,q} = [u_{i,q}^{(0)}, u_{i,q}^{(1)}]^T$, $\mathbf{z}_q = [z_q^{(0)}, z_q^{(1)}]^T$, $\mathbf{c}_i = [c_i^{(0)}, c_i^{(1)}]^T$ with

$$c_i^{(\tau)} = \cos^{(n_\tau+m+2)} \theta_i, \quad \tau = 0, 1, \quad (10)$$

and $\mathbf{k} = \begin{bmatrix} k_0 & 0 \\ 0 & k_1 \end{bmatrix}$ is a 2×2 matrix, where

$$\begin{aligned} k_\tau &= \frac{(n_\tau + 1) \beta \ell P_L A_U \mathbf{F}_i [1 + \rho(S_i, U, E_N, n_\tau)]}{2\pi h^2} \\ &= \kappa_\tau + \frac{H_{\text{NLOS}}(S_i, U, E_N, n_\tau)}{c_i^{(\tau)}}, \quad \tau = 0, 1, \end{aligned} \quad (11)$$

with κ_τ being a constant given by

$$\kappa_\tau = \frac{(n_\tau + 1) \beta \ell P_L A_U \mathbf{F}_i}{2\pi h^2}, \quad \tau = 0, 1. \quad (12)$$

Then, by dividing the RSS values corresponding to the two LEDs with respect to (8), we define the RSSR associated with the i -th DM-LED lamp as

$$\begin{aligned} \alpha_i &= \frac{u_{i,q}^{(\tau)}|_{\tau=0}}{u_{i,q}^{(\tau)}|_{\tau=1}} \\ &\approx \frac{n_0 + 1}{n_1 + 1} \cdot \frac{1 + \rho(S_i, U, E_N, n_0)}{1 + \rho(S_i, U, E_N, n_1)} \cdot \cos^{(n_0-n_1)} \theta_i \\ &= \frac{1}{\eta} \cdot \Gamma(S_i, U, E_N, n_0, n_1) \cdot \cos^{(n_0-n_1)} \theta_i, \end{aligned} \quad (13)$$

where $\eta = \frac{n_0+1}{n_1+1}$ and we have exploited $z_\tau \approx 0$ ($\tau = 0, 1$) assuming typical signal-to-noise ratios (SNR) in indoor scenarios [3], while

$$\Gamma(S_i, U, E_N, n_0, n_1) = \frac{1 + \rho(S_i, U, E_N, n_0)}{1 + \rho(S_i, U, E_N, n_1)} \quad (14)$$

is the channel gain ratio factor between n_0 and n_1 .

Based on (13), we arrive at

$$\begin{cases} \cos \hat{\theta}_i = \left[\frac{\alpha_i \eta}{\Gamma(S_i, U, E_N, n_0, n_1)} \right]^{\frac{1}{n_0-n_1}} \\ \hat{\theta}_i = \arccos \left[\frac{\alpha_i \eta}{\Gamma(S_i, U, E_N, n_0, n_1)} \right]^{\frac{1}{n_0-n_1}}, \end{cases} \quad (15)$$

where $\hat{\theta}_i \in [0, \frac{\pi}{2}]$, $\alpha_i \in [0, \frac{1}{\eta} \Gamma(S_i, U, E_N, n_0, n_1)]$ and $n_0 \neq n_1$. Note that when only the LOS path is considered, namely $H_{\text{NLOS}}(S_i, U, E_N, n_\tau) = 0$, according to (7), (11), (14) and (15), we will obtain $\rho(S_i, U, E_N, n_\tau) = 0$, $k_\tau = \kappa_\tau$, $\Gamma(S_i, U, E_N, n_0, n_1) = 1$ and $\cos \hat{\theta}_i = (\alpha_i \eta)^{\frac{1}{n_0-n_1}}$, where $\tau = \{0, 1\}$.

D. Analysis of the Channel Gain Ratio Factor

According to (3), we aim to first estimate the radiation angle θ_i , then to obtain d_i , and finally to exploit the algorithms proposed in Section III for computing the UE's spatial coordinate $U = (x_U, y_U, h_U) \in \mathbb{R}^3$. However, to get the estimated radiation angle $\hat{\theta}_i$ by exploiting (15) requires the knowledge of α_i and $\Gamma(S_i, U, E_N, n_0, n_1)$ defined in (14). This results in an analytical dilemma, since the variable U to be obtained through $\hat{\theta}_i$ also appears as an input parameter of $\Gamma(S_i, U, E_N, n_0, n_1)$. Thus, we propose to decouple U from $\Gamma(S_i, U, E_N, n_0, n_1)$.

To illustrate how this can be achieved, let us consider a general room model shown in Fig. 2, where I_L refers to the inter-lamp distance. Then, note from (14) that $\Gamma(S_i, U, E_N, n_0, n_1)$ is related to $\rho(S_i, U, E_N, n_\tau)$, which is a random variable depending on $H_{\text{LOS}}(S_i, U, n_\tau)$ and $H_{\text{NLOS}}(S_i, U, E_N, n_\tau)$, as defined in (7). Specifically, $H_{\text{LOS}}(S_i, U, n_\tau)$ can be calculated by (6), while $H_{\text{NLOS}}(S_i, U, E_N, n_\tau)$ can be computed with the aid of the iterative site-based model [31], [32]. To elaborate further, the surfaces of the room may be divided into N micro-reflective areas of a unit size $\frac{1}{P} \times \frac{1}{P}$ (m^2), and

$$N = P^2 \Phi, \quad (16)$$

where $\Phi = 2(l_R w_R + l_R h_R + w_R h_R)$ (m^2) is the surface area of the room. To extend the definition of $H^{(k)}(\cdot)$ in (5) to a more general form, we define $H_{\mathcal{L}_e, \mathcal{L}_r}^{(0)}$ and $H_{\mathcal{L}_e, \mathcal{L}_r}^{(k)}$ ($k = 1, \dots, K$) as the LOS and the NLOS channel gains from $\mathcal{L}_e$ to $\mathcal{L}_r$, respectively, where $\mathcal{L}_e$ refers to the location of a DM-LED or a micro-reflective area, $\mathcal{L}_r$ refers to the location of a micro-reflective area or the UE, and we have omitted E_N and n_τ for notation brevity. Then, according to [32, (13)] and the convolution property of the Fourier transform [37, (4.56)], we have

$$H_{S_i, U}^{(k)} \approx \sum_{p=1}^N \varrho_{\varepsilon_p^{(1)}} H_{S_i, \varepsilon_p^{(1)}}^{(0)} H_{\varepsilon_p^{(1)}, U}^{(k-1)}, \quad (17)$$

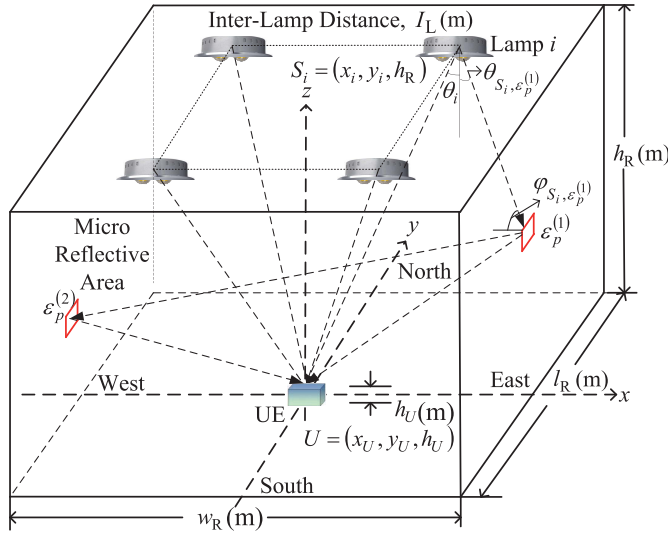


Fig. 2. A general VLP scenario considering multi-path propagations, where we set $K = 2$ as an example.

where

$$\begin{cases} H_{\varepsilon_p^{(1)},U}^{(k-1)} \approx \sum_{p=1}^N \varrho_{\varepsilon_p^{(2)}} H_{\varepsilon_p^{(1)},\varepsilon_p^{(2)}}^{(0)} H_{\varepsilon_p^{(2)},U}^{(k-2)} \\ \vdots \\ H_{\varepsilon_p^{(k-1)},U}^{(1)} \approx \sum_{p=1}^N \varrho_{\varepsilon_p^{(k)}} H_{\varepsilon_p^{(k-1)},\varepsilon_p^{(k)}}^{(0)} H_{\varepsilon_p^{(k)},U}^{(0)}, \end{cases} \quad (18)$$

while for notation convenience we let $\varepsilon_p^{(k)}$ denote both the p -th micro-reflective area and its location, which is passed by the light ray undergoing the k -th reflection, $p = 1, \dots, N$, $k = 1, \dots, K$, and $\varrho_{\varepsilon_p^{(k)}}$ is the reflection coefficients of $\varepsilon_p^{(k)}$.

Finally, combining (1), (3), (5)-(7), (14), (16) and (17), $\Gamma(S_i, U, E_N, n_0, n_1)$ of (14) may be approximated as

$$\begin{aligned} \Gamma(S_i, U, E_N, n_0, n_1) &= \frac{1 + \rho(S_i, U, E_N, n_0)}{1 + \rho(S_i, U, E_N, n_1)} \approx \frac{\frac{1}{\chi_c} + \frac{\sqrt{\pi} \gamma(\frac{n_0+1}{2})}{2\gamma(\frac{n_0}{2}+1) \cdot \cos(n_0+m+2) \theta_i}}{\frac{1}{\chi_c} + \frac{\sqrt{\pi} \gamma(\frac{n_1+1}{2})}{2\gamma(\frac{n_1}{2}+1) \cdot \cos(n_1+m+2) \theta_i}}, \end{aligned} \quad (19)$$

where $\chi_c > 0$ is a tuning factor independent of n_τ ($\tau = 0, 1$), and $\gamma(x) = \int_0^{+\infty} \nu^{x-1} e^{-\nu} d\nu$ is the standard Gamma function [38] that can be calculated by the MATLAB® function `gamma(·)`. More details on the derivation of $\Gamma(S_i, U, E_N, n_0, n_1)$ are provided in Appendix A.

To validate (19), we exploit (7) and (14) to plot the values of $\rho(S_i, U, E_N, n_\tau)$ and $\Gamma(S_i, U, E_N, n_0, n_1)$, respectively, at different combinations of $\{S_i, U\}$ with various configurations of E_N based on the iterative site-based model [31], [32], as shown in Fig. 3. More specifically, we vary the UE's location U across the entire room, along with different choices of S_i under two representative environmental configurations, namely $E_N = \{4 \text{ m}, 4 \text{ m}, 3 \text{ m}, 0.2 \text{ m}, 0.9, 0.9, 0.9, 0.9, 0.9, 0.5\}$ and

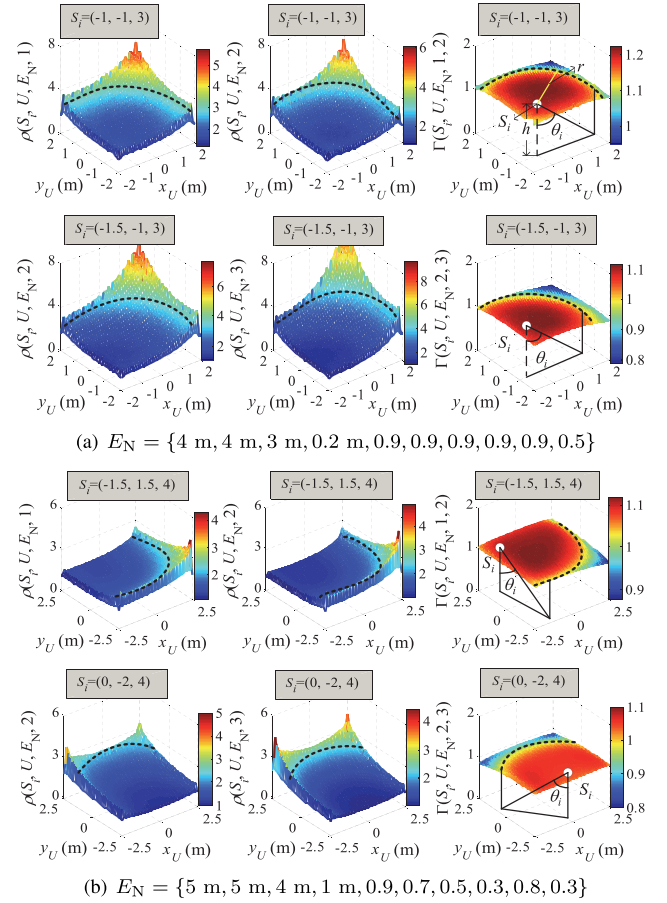


Fig. 3. Investigation of parameters $\rho(S_i, U, E_N, n_\tau)$ and $\Gamma(S_i, U, E_N, n_0, n_1)$ under hostile reflective environments, where $m = 1$.

$E_N = \{5 \text{ m}, 5 \text{ m}, 4 \text{ m}, 1 \text{ m}, 0.9, 0.7, 0.5, 0.3, 0.8, 0.3\}$ for homogeneously and heterogeneously reflective environments, respectively. Note that both are considered as hostile, heavily reflective environments.

From the results of Fig. 3, we note that the values of $\rho(S_i, U, E_N, 1)$ and $\rho(S_i, U, E_N, 2)$ have fluctuations under different combinations of $\{S_i, U\}$. However, they appear to be approximately similar within a large area of the room, which is constrained by a specific value range of θ_i associated with the LED lamps, as indicated by the dashed-line circles seen in Fig. 3. This interesting phenomenon implies that $\rho(S_i, U, E_N, 1) \approx \rho(S_i, U, E_N, 2)$ holds for a certain values of θ_i . Thus, according to (14), it is expected that $\Gamma(S_i, U, E_N, n_0, n_1)$ would have a value close to unity. This argument can be verified by the red 'plane-like' distributions of $\Gamma(S_i, U, E_N, 1, 2)$ and $\Gamma(S_i, U, E_N, 2, 3)$ in the four right-most subfigures within Fig. 3(a) and Fig. 3(b), where we can see the values of $\Gamma(S_i, U, E_N, n_0, n_1)$, $\{n_0, n_1\} \in \{\{1, 2\}, \{2, 3\}\}$ are indeed approximately equal to unity in a small-to-medium range of θ_i .

Aiming to verify the above analysis, we further plot the values of $\Gamma(S_i, U, E_N, n_0, n_1)$ measured under a range of θ_i in Fig. 4 based on (19), under given values of χ_c at different orders. Note from Appendix A that χ_c is a non-negative real-valued random variable related to the reflection coefficients of

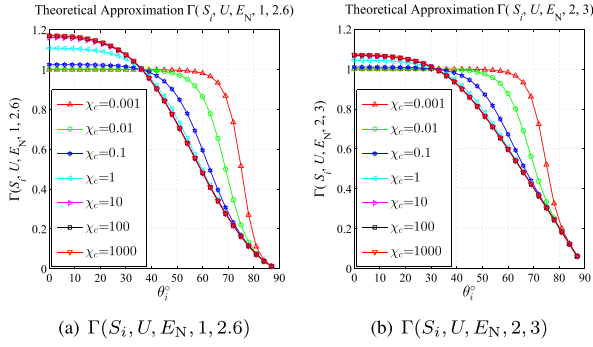


Fig. 4. The theoretical approximation of $\Gamma(S_i, U, E_N, n_0, n_1)$, where $m = 1$.

the micro-reflective areas. Since χ_c can take different values depending on the room environment, it is also beneficial to find out its potential impact on $\Gamma(S_i, U, E_N, n_0, n_1)$. Fig. 4 shows that within a small-to-medium range of θ_i , say for instance $\theta_i \in [0^\circ, \theta_T]$, where $\theta_T \in [30^\circ, 40^\circ]$, $\Gamma(S_i, U, E_N, n_0, n_1)$ approximately maintains a value close to unity regardless of the choice of χ_c . After extensive simulations under various configurations of E_N , we arrive at an important conclusion of

$$\Gamma(S_i, U, E_N, n_0, n_1) \approx \Gamma(n_0, n_1), \quad (20)$$

where $\Gamma(n_0, n_1)$ takes approximately constant values fluctuating around unity under some given range of θ_i , and is irrelevant to the specific values of $\{S_i, U\}$ and E_N .

Remark 1: We point out that utilizing the iterative site-based model of (17) to calculate $\Gamma(S_i, U, E_N, n_0, n_1)$ yields a very high complexity. Furthermore, it is difficult to obtain the value of ϱ_{ε_p} for each and every micro-reflective area ε_p ($p = 1, 2, \dots, N$) in any practical scenario. How to obtain $H_{\text{NLOS}}(S_i, U, E_N, n_\tau)$ in real environments remains to be an open issue in the literature. However, with the aid of the channel gain ratio factor $\Gamma(S_i, U, E_N, n_0, n_1)$, which approximately takes a near-unity constant value, we are capable of achieving an accurate positioning performance, taking into account $H_{\text{NLOS}}(S_i, U, E_N, n_\tau)$, but without the need to know its specific value.

III. POSITIONING ALGORITHMS EXPLOITING MULTIPLE DM-LED LAMPS

A. DM-LED Aided Non-Linear Least Squares (DM-NLS) Algorithm

Based on solid geometry theory, the cosine of θ_i shown in Fig. 1 can be calculated by

$$\cos \theta_i = \frac{h}{\sqrt{(x_i - x_U)^2 + (y_i - y_U)^2 + h^2}}. \quad (21)$$

By substituting (20) into (15) and exploiting (21), we define

$$\begin{aligned} \Delta_{\cos \theta_i} &= |\cos \theta_i - \cos \hat{\theta}_i| \\ &\approx \left| \frac{h}{\sqrt{(x_i - x_U)^2 + (y_i - y_U)^2 + h^2}} - \left[\frac{\alpha_i \eta}{\Gamma(n_0, n_1)} \right]^{\frac{1}{n_0 - n_1}} \right|, \end{aligned} \quad (22)$$

where we have replaced $\Gamma(S_i, U, E_N, n_0, n_1)$ in (15) with $\Gamma(n_0, n_1)$ according to (20), and η is defined in (13). Note that (22) has three unknown variables, x_U , y_U and $\Gamma(n_0, n_1)$. This implies that $V \geq 3$ DM-LED lamps, corresponding to three or more measured radiation angles, have to be detected by the PD to form an equation group for solving (22). In this case, it is equivalent to solve the following problem

$$\begin{aligned} &(\hat{x}_U, \hat{y}_U) \\ &= \arg \min_{(x_U, y_U)} \sum_{i=0}^{V-1} \left\{ \frac{h}{\sqrt{(x_i - x_U)^2 + (y_i - y_U)^2 + h^2}} - \left[\frac{\alpha_i \eta}{\Gamma(n_0, n_1)} \right]^{\frac{1}{n_0 - n_1}} \right\}^2 \\ &s.t. \begin{cases} X_{\min} \leq x_U \leq X_{\max} \\ Y_{\min} \leq y_U \leq Y_{\max} \\ \Gamma_{\min} \leq \Gamma(n_0, n_1) \leq \Gamma_{\max} \end{cases}, \end{aligned} \quad (23)$$

where $(\cdot)_{\min}$ and $(\cdot)_{\max}$ denote the minimal and maximal values of the corresponding variable, respectively.

The solution to (23) can be obtained through the interior point method of the MATLAB® function `fmincon`, where the output local minima depends on $\Gamma(n_0, n_1)$ and the initial values of (x_U, y_U) . To determine the ranges and initial values of the constrained parameters in (23), we assume that the coordinates of $V \geq 3$ DM-LED lamps are known at the UE. Although how to obtain the coordinates of the lamps is out of the scope of this work, we point out that there exist a number of LED-ID detection methods [25]–[27] that can help achieve this objective. Then, three out of the V coordinates are selected, which correspond to the three lamps that yield the three strongest RSSs. Next, we define

$$\begin{cases} X_{\min} = \min \{x_{E,1}, x_{E,2}, x_{E,3}\} \\ X_{\max} = \max \{x_{E,1}, x_{E,2}, x_{E,3}\} \\ Y_{\min} = \min \{y_{E,1}, y_{E,2}, y_{E,3}\} \\ Y_{\max} = \max \{y_{E,1}, y_{E,2}, y_{E,3}\}, \end{cases} \quad (24)$$

where $(x_{E,i}, y_{E,i})$ ($i = 1, 2, 3$) represents the coordinate of the i -th lamp. Then, the initial values of x_U and y_U in (23), denoted as $x_U^{(0)}$ and $y_U^{(0)}$, may be set to the centroid of the virtual rectangle area bounded by the four lines of $x = X_{\min}$, $x = X_{\max}$, $y = Y_{\min}$ and $y = Y_{\max}$ in the coordinate system, yielding

$$\begin{cases} x_U^{(0)} = \frac{1}{2}(X_{\min} + X_{\max}) \\ y_U^{(0)} = \frac{1}{2}(Y_{\min} + Y_{\max}). \end{cases} \quad (25)$$

On the other hand, the range of $\Gamma(n_0, n_1)$ may be empirically determined according to simulation results as those illustrated in Fig. 4. For example, the range of $\Gamma(n_0, n_1) \in [\Gamma_{\min}, \Gamma_{\max}] = [0.5, 1.5]$ may be used.

B. DM-LED Aided Linear Least Squares (DM-LLS) Algorithm

To avoid the local minimum searching procedure required by the DM-NLS approach of Section III-A, we may transform

the second-order formation of (23) to its first order. Firstly, the optimization objective will be to make (22) equal to zero, yielding

$$\frac{h}{\sqrt{(x_i - x_U)^2 + (y_i - y_U)^2 + h^2}} = \left[\frac{\alpha_i \eta}{\Gamma(n_0, n_1)} \right]^{\frac{1}{n_0 - n_1}}. \quad (26)$$

Then, define $\bar{\Gamma} = \left[\frac{\Gamma(n_0, n_1)}{\eta} \right]^{\frac{2}{n_0 - n_1}}$ and $\mathfrak{R}_i = \frac{2}{\alpha_i^{n_1 - n_0}}$. According to the discussions on (20) in Section II-D, $\Gamma(n_0, n_1)$ may be approximated as a constant. Thus, $\bar{\Gamma}$ may also be considered as a constant, and (26) can be rewritten to

$$(x_i - x_U)^2 + (y_i - y_U)^2 + h^2 = h^2 \bar{\Gamma} \mathfrak{R}_i. \quad (27)$$

Again, assuming that the UE detects signals transmitted from $V \geq 3$ DM-LED lamps, we get a group of equations from (27) as

$$\begin{cases} (x_U - x_0)^2 + (y_U - y_0)^2 + h^2 = h^2 \bar{\Gamma} \mathfrak{R}_0 \\ (x_U - x_1)^2 + (y_U - y_1)^2 + h^2 = h^2 \bar{\Gamma} \mathfrak{R}_1 \\ \vdots \\ (x_U - x_{V-1})^2 + (y_U - y_{V-1})^2 + h^2 = h^2 \bar{\Gamma} \mathfrak{R}_{V-1}. \end{cases} \quad (28)$$

Furthermore, by invoking the subtraction operation on every two equations in (28), we obtain a number of $\frac{V(V-1)}{2}$ equations, which can be expressed as

$$\mathbf{A}\mathbf{g} = \mathbf{D}, \quad (29)$$

where $\mathbf{g} = [x_U, y_U]^T$, and

$$\mathbf{A} = \begin{bmatrix} 2(x_0 - x_1) & 2(y_0 - y_1) \\ \vdots & \vdots \\ 2(x_0 - x_{V-1}) & 2(y_0 - y_{V-1}) \\ 2(x_1 - x_2) & 2(y_1 - y_2) \\ \vdots & \vdots \\ 2(x_1 - x_{V-1}) & 2(y_1 - y_{V-1}) \\ \vdots & \vdots \\ 2(x_{V-2} - x_{V-1}) & 2(y_{V-2} - y_{V-1}) \end{bmatrix}, \quad (30)$$

$$\mathbf{D} = \begin{bmatrix} x_0^2 + y_0^2 - x_1^2 - y_1^2 + h^2 \bar{\Gamma} (\mathfrak{R}_1 - \mathfrak{R}_0) \\ \vdots \\ x_0^2 + y_0^2 - x_{V-1}^2 - y_{V-1}^2 + h^2 \bar{\Gamma} (\mathfrak{R}_{V-1} - \mathfrak{R}_0) \\ x_1^2 + y_1^2 - x_2^2 - y_2^2 + h^2 \bar{\Gamma} (\mathfrak{R}_2 - \mathfrak{R}_1) \\ \vdots \\ x_1^2 + y_1^2 - x_{V-1}^2 - y_{V-1}^2 + h^2 \bar{\Gamma} (\mathfrak{R}_{V-1} - \mathfrak{R}_1) \\ \vdots \\ x_{V-2}^2 + y_{V-2}^2 - x_{V-1}^2 - y_{V-1}^2 + h^2 \bar{\Gamma} (\mathfrak{R}_{V-1} - \mathfrak{R}_{V-2}) \end{bmatrix}. \quad (31)$$

Then, we may solve (29) by the linear least squares algorithm as

$$[\hat{x}_U, \hat{y}_U] = \hat{\mathbf{g}}^T = \mathbf{D}^T \mathbf{A} (\mathbf{A}^T \mathbf{A})^{-1}. \quad (32)$$

Remark 2: It is worth pointing out that the conventional RSS-nVLP aided linear least squares (RSS-LLS) [13] and the proposed DM-LLS schemes have a similarly low complexity,

as they exploit the same set of linear equations in the form of (32), despite the different element values of $\mathbf{D}$. The proposed DM-NLS scheme yields a higher complexity, but typically achieves a better performance than DM-LLS and RSS-LLS, as to be demonstrated in Section V.

IV. THE ANALYSIS OF CRAMÉR-RAO LOWER BOUND

The CRLB is a bound on the variance of unbiased estimators, which provides a useful hint about the achievable lower bound of the mean squared error (MSE) performance of the target estimator, namely the estimate of d_i in our case, denoted by $\hat{d}_i$. With respect to the DM-nVLP model of Fig. 1, when we have a more accurate $\hat{d}_i$ for (3), the reliability of the estimated UE location also improves. In this section, we analyze the CRLB of $\hat{d}_i$ for evaluating the impact on distance estimation imposed by the noise $z_q^{(\tau)}$ of (8).

Define $\mathbf{u}_i = [\mathbf{u}_{i,1}, \mathbf{u}_{i,2}, \dots, \mathbf{u}_{i,Q}]$, where $\mathbf{u}_{i,q}$ ($i = 0, \dots, V-1$; $q = 0, \dots, Q-1$) is given in (9). According to [39], the conditional joint probability density function (PDF) of $\mathbf{z}_q = \mathbf{u}_{i,q} - \mathbf{k}c_i$ is given by

$$\begin{aligned} p(\mathbf{u}_i; \mathbf{c}_i) &= \prod_{q=0}^{Q-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} [u_{i,q}^{(0)} - k_0 c_i^{(0)}]^2 \right\} \\ &\quad \cdot \prod_{q=0}^{Q-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} [u_{i,q}^{(1)} - k_1 c_i^{(1)}]^2 \right\} \\ &= \frac{1}{(2\pi\sigma^2)^{\frac{Q}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{q=0}^{Q-1} [u_{i,q}^{(0)} - k_0 c_i^{(0)}]^2 \right\} \\ &\quad \cdot \frac{1}{(2\pi\sigma^2)^{\frac{Q}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{q=0}^{Q-1} [u_{i,q}^{(1)} - k_1 c_i^{(1)}]^2 \right\} \\ &= \frac{1}{(2\pi\sigma^2)^Q} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{q=0}^{Q-1} \left[(z_q^{(0)})^2 + (z_q^{(1)})^2 \right] \right\} \end{aligned} \quad (33)$$

Since the PDF of $\mathbf{z}_q$, denoted by $p(\mathbf{u}_i; \mathbf{c}_i)$, satisfies the regularity condition [39], the CRLB of the estimator $\hat{c}_i$ always exists. Furthermore, the log-likelihood function corresponding to (33) can be written as

$$\begin{aligned} \ln p(\mathbf{u}_i; \mathbf{c}_i) &= -Q \ln 2\pi - Q \ln \sigma^2 \\ &\quad - \frac{1}{2\sigma^2} \sum_{q=0}^{Q-1} [u_{i,q}^{(0)} - k_0 c_i^{(0)}]^2 \\ &\quad - \frac{1}{2\sigma^2} \sum_{q=0}^{Q-1} [u_{i,q}^{(1)} - k_1 c_i^{(1)}]^2. \end{aligned} \quad (34)$$

Based on the classical signal estimation theory [39], the CRLB of the parameter vector $\mathbf{c}_i = [c_i^{(0)}, c_i^{(1)}]^T$ can be obtained by exploiting the 2×2 Fisher information matrix, which is calculated by

$$\mathbf{M}(\mathbf{c}_i) = \begin{bmatrix} -E \left\{ \frac{\partial^2 \ln p(\mathbf{u}_i; \mathbf{c}_i)}{\partial (c_i^{(0)})^2} \right\} & -E \left\{ \frac{\partial^2 \ln p(\mathbf{u}_i; \mathbf{c}_i)}{\partial c_i^{(0)} \partial c_i^{(1)}} \right\} \\ -E \left\{ \frac{\partial^2 \ln p(\mathbf{u}_i; \mathbf{c}_i)}{\partial c_i^{(1)} \partial c_i^{(0)}} \right\} & -E \left\{ \frac{\partial^2 \ln p(\mathbf{u}_i; \mathbf{c}_i)}{\partial (c_i^{(1)})^2} \right\} \end{bmatrix}, \quad (35)$$

where $E\{\cdot\}$ is the expectation operation. The Fisher information matrix given by (35) involves partial derivatives of (34), which further exploit the derivatives of k_τ ($\tau = 0, 1$) defined in (11). Note that k_τ contains the unknown variables $H_{\text{NLOS}}(S_i, U, E_N, n_\tau)$, which, as discussed at the end of Section II-B, may be treated as some constants associated with the specific combination of S_i and U under a given E_N . Based on this assumption, we may therefore simplify (35) with the aid of (11) and (34), to

$$\mathbf{M}(\mathbf{c}_i) = \begin{bmatrix} \frac{Q\kappa_0^2}{\sigma^2} & 0 \\ 0 & \frac{Q\kappa_1^2}{\sigma^2} \end{bmatrix}. \quad (36)$$

Then, to estimate d_i , we combine (3) and (10) to yield

$$d_i = h \left[\frac{c_i^{(1)}}{c_i^{(0)}} \right]^{\frac{1}{n_0 - n_1}} = g(\mathbf{c}_i), \quad (37)$$

where $g(\cdot)$ is defined as a function of the vector parameter $\mathbf{c}_i$. According to [39, Section 3.8], to calculate the CRLB of $\hat{d}_i$, we need the Jacobian matrix related to $\mathbf{c}_i$, which is given by

$$\begin{aligned} & \frac{\partial g(\mathbf{c}_i)}{\partial \mathbf{c}_i} \\ &= \begin{bmatrix} \frac{\partial g(\mathbf{c}_i)}{\partial c_i^{(0)}}, \frac{\partial g(\mathbf{c}_i)}{\partial c_i^{(1)}} \end{bmatrix} \\ &= \begin{bmatrix} h(c_i^{(0)})^{\frac{1+n_0-n_1}{n_1-n_0}}, h(c_i^{(1)})^{\frac{1+n_1-n_0}{n_0-n_1}} \\ (n_1-n_0)(c_i^{(1)})^{\frac{1}{n_1-n_0}}, (n_0-n_1)(c_i^{(0)})^{\frac{1}{n_0-n_1}} \end{bmatrix}. \end{aligned} \quad (38)$$

By substituting (10) into (38), we have

$$\begin{cases} \frac{\partial g(\mathbf{c}_i)}{\partial c_i^{(0)}} = \frac{h}{n_1 - n_0} \cos^{-(n_0+m+3)} \theta_i \\ \frac{\partial g(\mathbf{c}_i)}{\partial c_i^{(1)}} = \frac{h}{n_0 - n_1} \cos^{-(n_1+m+3)} \theta_i. \end{cases} \quad (39)$$

On the other hand, according to [39, (3.30)], the covariance matrix of the estimator $\hat{d}_i$ satisfies

$$\mathbf{C}_{\hat{d}_i} \geq \mathbf{B}, \quad (40)$$

where by exploiting (36), we define

$$\begin{aligned} \mathbf{B} &= \left[\frac{\partial g(\mathbf{c}_i)}{\partial \mathbf{c}_i} \right] \cdot \mathbf{M}^{-1}(\mathbf{c}_i) \cdot \left[\frac{\partial g(\mathbf{c}_i)}{\partial \mathbf{c}_i} \right]^T \\ &= \begin{bmatrix} \frac{\partial g(\mathbf{c}_i)}{\partial c_i^{(0)}} & \frac{\partial g(\mathbf{c}_i)}{\partial c_i^{(1)}} \end{bmatrix} \cdot \begin{bmatrix} \frac{\sigma^2}{Q\kappa_0^2} & 0 \\ 0 & \frac{\sigma^2}{Q\kappa_1^2} \end{bmatrix} \cdot \begin{bmatrix} \frac{\partial g(\mathbf{c}_i)}{\partial c_i^{(0)}} \\ \frac{\partial g(\mathbf{c}_i)}{\partial c_i^{(1)}} \end{bmatrix} \\ &= \frac{\sigma^2}{Q} \left[\frac{1}{\kappa_0^2} \left(\frac{\partial g(\mathbf{c}_i)}{\partial c_i^{(0)}} \right)^2 + \frac{1}{\kappa_1^2} \left(\frac{\partial g(\mathbf{c}_i)}{\partial c_i^{(1)}} \right)^2 \right]. \end{aligned} \quad (41)$$

Then, by substituting (3), (12) and (39) into (41), we arrive at

$$\begin{aligned} \mathbf{B} &= \frac{4\pi^2\sigma^2}{\beta^2\ell^2 P_L^2 A_U^2 Q(n_1-n_0)^2} \cdot \sum_{\tau=0}^1 \frac{h^6}{(n_\tau+1)^2 \cos^{2(n_\tau+m+3)} \theta_i} \\ &= \frac{4\pi^2\sigma^2}{\beta^2\ell^2 P_L^2 A_U^2 Q(n_1-n_0)^2} \cdot \sum_{\tau=0}^1 \frac{d_i^{2(n_\tau+m+3)}}{(n_\tau+1)^2 h^{2(n_\tau+m)}}. \end{aligned} \quad (42)$$

Furthermore, since d_i is a scalar, $\mathbf{C}_{\hat{d}_i}$ in (40) becomes a variance, yielding

$$\mathbf{C}_{\hat{d}_i} = \text{var}_{\hat{d}_i} = E \left[\left(\hat{d}_i - d_i \right)^2 \right] \geq \mathbf{B}. \quad (43)$$

Note that the left-hand side of (43) also refers to the MSE of $\hat{d}_i$. Hence, by substituting (42) into (43), the CRLB of $\hat{d}_i$ under the proposed DM-nVLP model can be obtained by

$$\begin{aligned} \text{MSE}_{\hat{d}_i}^{\text{DM}} &= \text{var}_{\hat{d}_i} \\ &\geq \frac{4\pi^2\sigma^2}{\beta^2\ell^2 P_L^2 A_U^2 Q(n_1-n_0)^2} \cdot \sum_{\tau=0}^1 \frac{d_i^{2(n_\tau+m+3)}}{(n_\tau+1)^2 h^{2(n_\tau+m)}}, \end{aligned} \quad (44)$$

In order to minimize the CRLB under a given combination of $\{n_0, n_1\}$, we propose an objective function

$$\begin{aligned} & \{\hat{n}_0, \hat{n}_1\} \\ &= \arg \min_{\{n_0, n_1\}} \left\{ \frac{4\pi^2\sigma^2}{\beta^2\ell^2 P_L^2 A_U^2 Q(n_1-n_0)^2} \cdot \sum_{\tau=0}^1 \frac{d_i^{2(n_\tau+m+3)}}{(n_\tau+1)^2 h^{2(n_\tau+m)}} \right\} \\ & \text{s.t. } n_0 \neq n_1; n_0 < n_1 \leq n_{\max}; n_0, n_1, n_{\max} \in \mathbb{R}^+, \end{aligned} \quad (45)$$

where $n_{\max}$ is the maximal RLMN of the LED. Given the parameter set of $\{\beta, \ell, A_U, m, P_L, d_i, h, \sigma^2, Q, n_{\max}\}$, the solution to (45) may be obtained offline through grid-based searching based methods. For example, assuming $\{\beta, \ell, A_U, m, P_L, d_i, h, \sigma^2, Q, n_{\max}\} = \{0.4 \text{ A/W}, 10, 1 \text{ cm}^2, 1, 2 \text{ W}, 5 \text{ m}, 3 \text{ m}, 10^{-13} \text{ A}^2, 1, 10\}$, one local optimum RLMN pair minimizing (45) is $\{n_0, n_1\}_{\text{opt}} = \{1, 4.6\}$.

For benchmarking purpose and taking into account Q measurements for fairness, we also provide the CRLB of the conventional RSS-nVLP model described in [3] as

$$\text{MSE}_{\hat{d}_i}^{\text{SM}} \geq \frac{4\pi^2\sigma^2 d_i^{2(n+m+3)}}{(n+1)^2 (n+m+2)^2 \beta^2 \ell^2 P_S^2 A_U^2 Q h^{2(n+m)}}, \quad (46)$$

where P_S denotes the peak optical power of the single LED in the RSS system. For fair comparison between the conventional RSS and the proposed DM-LED schemes, we set $P_S = 2P_L$ in the sequel.

Remark 3: We point out that the CRLBs represented by (44) and (46) correspond to the ideal scenario, where they are assumed to be affected only by the LOS channel gain

TABLE I
THE MAIN PARAMETERS OF THE DM-nVLP SYSTEM

Parameter	Value	Parameter	Value
Lamp locations	S_0 (-0.6, -0.6, 2.33) m	A_U	1 cm ²
	S_1 (0.6, -0.6, 2.33) m	β	0.4 A/W
	S_2 (-0.6, 0.6, 2.33) m	FOV	120°
	S_3 (0.6, 0.6, 2.33) m	σ^2	10 ⁻¹³ A ² [4]
h_R	2.33 m	I_L	1.2 m
h_U	0.5 m	ℓ	10

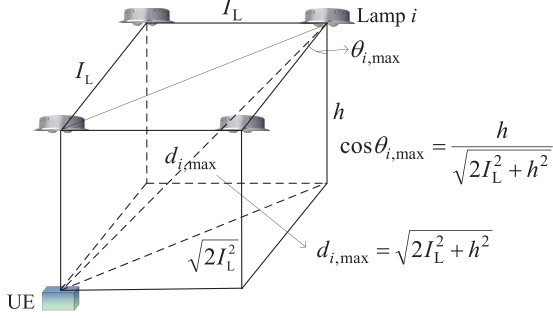


Fig. 5. The nVLP system employing a square array of LED lamps in a typical indoor scenario.

and AWGN, without the impact from the NLOS channel gain. This is a result due to the fact that (44) and (46) are derived by approximating $H_{\text{NLOS}}(S_i, U, E_N, n_\tau)$ as a constant, such that the differential operations in (35) can be simplified. Nonetheless, in practical scenarios, the NLOS channel gain may vary and its impact cannot be ignored. We will revisit this issue in Section V-A.2.

V. EVALUATION RESULTS

A. Analytical and Simulation Results

In this section, we provide analytical and simulation results of the proposed DM-nVLP system, under the room model of Fig. 2 with multi-path reflections. We assume that the UE is capable of receiving signals from at least $V \geq 3$ DM-LEDs, indicating $\mathbf{F}_i = 1$ for $i = 0, \dots, V-1$. The main parameters take the values in Table I unless otherwise stated.

1) *CRLB Performance*: In this section, we compare the square-root CRLB (rCRLB) performances of the proposed DM-nVLP and the conventional RSS-nVLP systems under the theoretically idealized scenario, where a constant NLOS channel gain is assumed.

Observe (44) and (46) that the value of rCRLB increases when d_i becomes larger. Thus, based on the general room model of Fig. 5 employing a square array of LED lamps, we opt for evaluating the worse case, where $d_i = d_{i,\max} = \sqrt{2I_L^2 + h^2}$, which is the maximal distance between the i -th lamp and the UE. In this case, the i -th lamp has a maximal radiation angle $\theta_{i,\max}$ relative to the UE, resulting in

$$\cos \theta_{i,\max} = \frac{h}{\sqrt{h^2 + 2I_L^2}}. \quad (47)$$

Exploiting the parameters in Table I, we can obtain $d_{i,\max} = 2.496$ m and $\theta_{i,\max} = 42.85^\circ$.

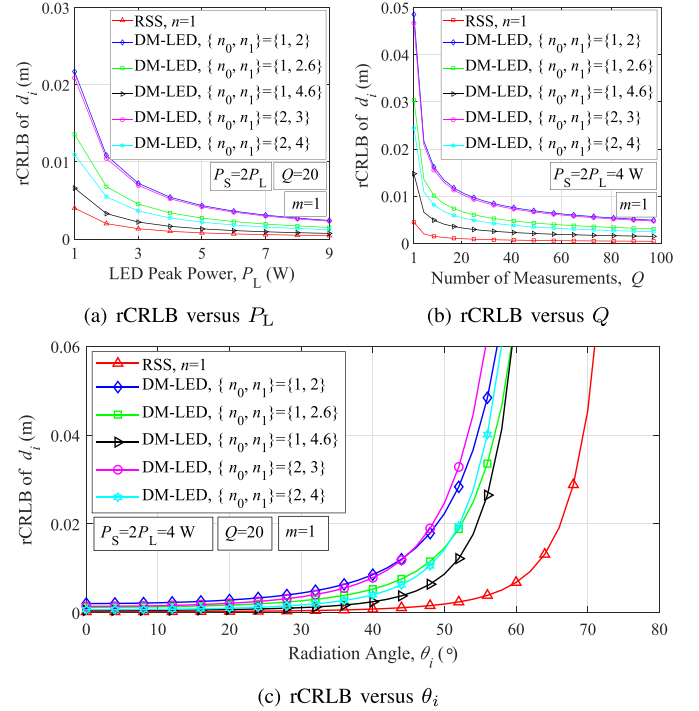


Fig. 6. The rCRLB calculated with various RLMN combinations under the ideal scenario assuming a constant NLOS channel gain.

Based on the above-mentioned assumptions, we can calculate the worst-case rCRLBs of the DM-LED and the RSS systems using (44) and (46), respectively. In Fig. 6(a), we plot the rCRLB values versus LED peak power P_L for both models assuming $Q = 20$ measurements, where we set $n = 1$ for RSS and various combinations of $\{n_0, n_1\}$ for DM-LED. It is observed that the rCRLB of DM-LED with various RLMN combinations is higher than that of RSS with a single RLMN. As P_L increases to a value beyond 4 W, the rCRLB of DM-LED reduces below 1 cm for all combinations of $\{n_0, n_1\}$ tested, approaching that of the RSS scheme. On the other hand, in Fig. 6(b) we show the rCRLB performance by varying Q under a fixed LED peak power of $P_L = 2$ W. Similar to Fig. 6(a), DM-LED is outperformed by RSS in terms of the rCRLB metric. However, the achievable performance of DM-LED can be improved at a larger Q , thanks to a better AWGN-smoothing effect. Furthermore, from Fig. 6(a) and Fig. 6(b), we can see that the combination of $\{n_0, n_1\} = \{1, 4.6\}$ yields the lowest rCRLB for DM-LED, which is consistent with the optimized result obtained through (45) based on the grid-based searching approach.

Next, we investigate the rCRLB performance versus the radiation angle θ_i in Fig. 6(c), where we set $P_L = 2$ W and $Q = 20$. Naturally, the value of θ_i is associated with the UE's specific location in the room. From Fig. 6(c), we can observe that the rCRLB of DM-LED with $\{n_0, n_1\} = \{1, 4.6\}$ approaches its best performance, which is very close to that of RSS-based scheme, when $\theta_i \in [0^\circ, \theta_T]$ with $\theta_T \in [30^\circ, 40^\circ]$. As long as $\theta_i \leq 50^\circ$, the rCRLBs of both DM-LED and RSS based schemes can be maintained below 3 cm, although they deteriorate quickly when θ_i becomes larger. The results in Fig. 6(c) provide useful hints for designing the lamp

TABLE II
ENVIRONMENTAL PARAMETER CONFIGURATIONS

Scenario	Environmental parameter set, E_N
Zero reflection	{1.8 m, 1.8 m, 2.33 m, 0.5 m, 0, 0, 0, 0, 0}
Corridor reflection	{1.8 m, 1.8 m, 2.33 m, 0.5 m, 0.8, 0.8, 0, 0, 0.6, 0.3}
Corner reflection	{1.8 m, 1.8 m, 2.33 m, 0.5 m, 0.8, 0, 0, 0.8, 0.6, 0.3}
Surround reflection	{1.8 m, 1.8 m, 2.33 m, 0.5 m, 0.8, 0.8, 0.8, 0.8, 0.8, 0.4}

deployment scheme. For instance, given a targeted positioning accuracy, the maximal θ_i can be decided according to Fig. 6(c), which helps to derive the recommended inter-lamp distance I_L based on (47).

Remark 4: Note that the results in Fig. 6 were generated under the ideal scenario discussed at the end of Section IV, which only revealed the attainable CRLB performances under LOS channel gain and AWGN, assuming constant NLOS channel gain. However, when a varying NLOS channel gain is taken into account, the conventional RSS-nVLP system suffers a significant performance degradation, as to be shown in Section V-A.2.

2) Performance of the Proposed Positioning Algorithms:

The NLOS channel gain $H_{\text{NLOS}}(S_i, U, E_N, n_\tau)$ is not negligible in realistic scenarios [31], [32], especially when the UE is close to the walls, corners, furniture and even when people walk in the room. In order to achieve a high positioning accuracy, theoretically, the NLOS channel gain associated with the combination of any DM-LED lamp and any location in the room, $\{S_i, U\}$, has to be acquired. Nevertheless, it is very challenging to measure all instances of $\{S_i, U\}$ in practical VLP scenarios. In addition, if the environment changes due to, for example furniture rearrangement and people's movement, the reflection topology will be changed and the NLOS channel gain has to be remeasured in real-time. In fact, $H_{\text{NLOS}}(S_i, U, E_N, n_\tau)$ is a random variable with unknown value due to the environmental uncertainty. Since the conventional RSS-nVLP system cannot cope with the NLOS channel gain, its performance substantially degrades under a reflective environment, as pointed out by [13] and by Fig. 7.

More specifically, Fig. 7 shows simulation results in an idealized reflection-free and a few scenarios with corridor, corner and surround reflections, where the corresponding configurations of E_N are outlined in Table II. We adopt the evaluation metric of location error (LE), defined by $\sqrt{(x_U - \hat{x}_U)^2 + (y_U - \hat{y}_U)^2}$, where $(\hat{x}_U, \hat{y}_U)$ denotes the estimated coordinate of the UE location. In the DM-NLS algorithm, we obtain $(\hat{x}_U, \hat{y}_U)$ through (23), subjected to the constraints of $[X_{\min}, X_{\max}] = [Y_{\min}, Y_{\max}] = [-0.6, 0.6]$, $[\Gamma_{\min}, \Gamma_{\max}] = [0.1, 2]$, and $(x_U^{(0)}, y_U^{(0)}) = (0, 0)$ based on (25). In the DM-LLS algorithm, we apply (29) and set $\Gamma(n_0, n_1) = 1.1$. For comparison, the RSS-LLS algorithm [13] is selected as a benchmark, though the RSS-nVLP aided nonlinear least squares (RSS-NLS) algorithm is not shown due to its even worse performance.

In the idealized zero-reflection case of Fig. 7(a), as expected, our observations comply with those made in Section V-A.1, where the DM-LED model is outperformed by its RSS counterpart under LOS propagation. However, both DM-LED schemes substantially outperform the RSS scheme under

NLOS propagation scenarios, reducing the average LE by about 50-80% across the square region projected by the four DM-LED lamps. Furthermore, more performance gains are achievable when the NLOS reflection becomes severer from Fig. 7(b) to Fig. 7(d). This is expected, since the DM-LED model takes into account the NLOS channel gain with the aid of $\Gamma(S_i, U, E_N, n_0, n_1)$, as mentioned at the end of Section II-D, thus effectively improves the achievable performance. Most importantly, the DM-LED model does not require the value of $H_{\text{NLOS}}(S_i, U, E_N, n_0, n_1)$, as it approximately decouples the correlation between multi-path reflections and the UE's specific location, which makes the implementation of high-performance nVLP solutions practically feasible. In addition, an average LE performance of about 4-9 cm can be obtained by the DM-LED schemes, in all scenarios of Fig. 7 regardless of the level of reflection. This is practically beneficial for maintaining a stable quality of experience for users.

B. Experimental Results

1) *Prototype DM-LED Lamps:* To our best knowledge, there is no DM-LED lamp available on the current market. To validate the DM-nVLP model, we manually design and implement the first ever prototype DM-LED lamp. In our design example, the lamp employs two types of OSRAM® LED chips with different RLMNs, namely GW PUSRA1.PM (OSL-120) [40] and GW CS8PM1.PM (OSL-80) [41], which have a rated power of 2 W and 1 W, respectively, as shown in Fig. 8(a). OSL-120 is a standard Lambertian emitter with a half-power viewing angle of 120° [40], namely $\theta_{\frac{1}{2},0} = 60^\circ$, while OSL-80 has a half-power viewing angle of about 80° [41], namely $\theta_{\frac{1}{2},1} \approx 40^\circ$. Based on (2), the RLMNs of the OSL-120 and OSL-80 are $n_0 = 1$ and $n_1 \approx 2.6$, respectively. After a careful mounting of both chips onto the heat sink, a gap of $d_I = 2$ cm remains between the two LED chips due to the relatively larger size of the aluminium substrates. Recall from Section II-A that we have the ideal assumption of $d_I \approx 0$ when deriving the theoretical DM-nVLP system model. Naturally, the non-ideality of d_I would have some impact on the system's achievable performance. However, we point out that it does not affect our conclusions later.

Fig. 8(b) and Fig. 8(c) show the normalized light intensities calculated by $I'(\theta_i, n_\tau) = \frac{2\pi}{n_\tau + 1} I(\theta_i, n_\tau)$, $\tau = \{0, 1\}$, where $I(\theta_i, n_\tau)$ is defined in (1), versus the radiation angle θ_i for $n_0 = 1$ and $n_1 = 2.6$, respectively, and compare them with the normalized intensities of OSL-120 [40] and OSL-80 [41], respectively. From Fig. 8(b), we note that since OSL-120 conforms with the normalized standard Lambertian radiation with $n_0 = 1$, the official curve given in [40] can well match the normalized intensity $I'(\theta_i, n_0 | n_0 = 1)$ derived from (1) in the full range of $\theta_i \in [0^\circ, 90^\circ]$. On the other hand, the official curve of the non-Lambertian-radiation chip OSL-80 [41] matches the derived result under $n_1 = 2.6$ within only the partial range of $\theta_i \in [0^\circ, 45^\circ]$, as seen in Fig. 8(c).

2) *Experimental Setup:* The basic experimental setup of the prototype DM-nVLP system is shown in Fig. 9, where we used the eight out of nine channels from three programmable direct

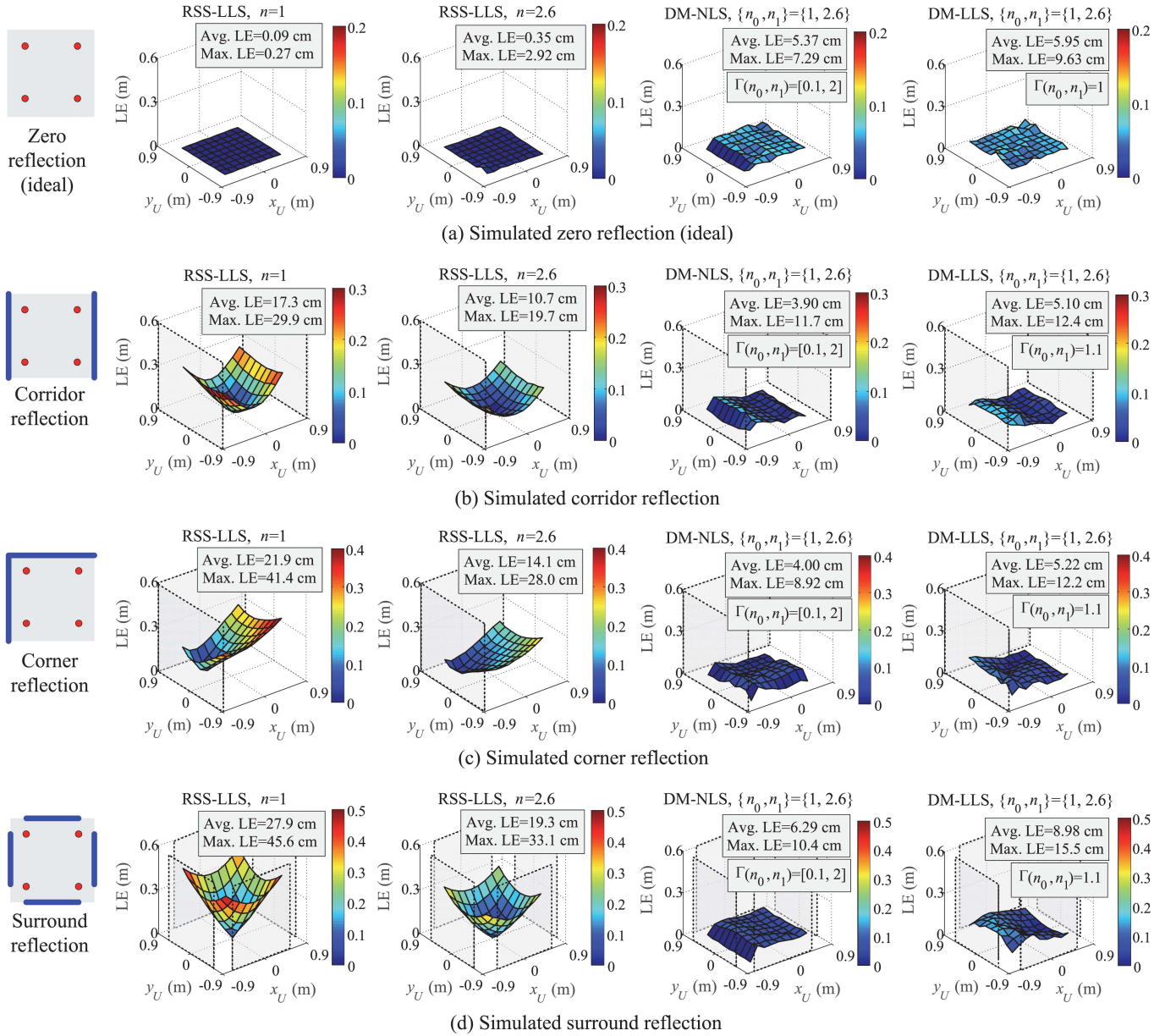


Fig. 7. The LE performance comparisons of DM-LED and RSS-nVLP systems simulated with $P_S = 2P_L = 4$ W, $m = 1$, $d_I = 2$ cm, and $\{n_0, n_1\} = \{1, 2.6\}$.

current power supplies, RIGOL DP832, to support eight LED chips of four DM-LED lamps. The lamps were equidistantly mounted on the ceiling with an inter-lamp distance of $I_L = 1.2$ m. An STM32F10 development board was used to generate the pulse width modulation signals. An N -channel metal oxide semiconductor field effect transistor, FQD60N03L, was exploited to construct an on-off switching circuit for flickering control of each LED chip on a predefined square waveform at frequency $f_{i,\tau}$. The frequency configuration for the DM-LED lamps is outlined in Table III.

At the receiver side, we used a silicon pin PD, 2DU10, which has a receiving area of 1 cm^2 , as seen in Fig. 9. The signal received by the PD was forwarded to a laptop's microphone port through a 3.5 mm audio cable, such that we could exploit the laptop's built-in sound card to amplify and then convert the optical signal to its digital version,

TABLE III
FREQUENCY CONFIGURATION FOR THE DM-LED LAMPS

	LED-0: OSL-120 (kHz)	LED-1: OSL-80 (kHz)
DM-LED 0	$f_{0,0} = 2.3$	$f_{0,1} = 1.4$
DM-LED 1	$f_{1,0} = 2.5$	$f_{1,1} = 1.6$
DM-LED 2	$f_{2,0} = 2.7$	$f_{2,1} = 1.8$
DM-LED 3	$f_{3,0} = 2.9$	$f_{3,1} = 2.0$

thus eliminating the need to design an independent signal amplifier and analog-to-digital convertor. Then, the National Instruments™ LabVIEW sound acquisition functions were used to obtain the digital signal data. Furthermore, the Goertzel algorithm was utilized to generate the spectral amplitude of the square waveforms transmitted from each LED of each DM-LED lamp. Aiming to mitigate interferences between different

TABLE IV
THE AVERAGE LES REDUCED FROM RSS-nVLP TO DM-nVLP SYSTEMS

Scenario	LE reduction from RSS-nVLP (OSL-120)		LE reduction from RSS-nVLP (OSL-80)	
	DM-NLS	DM-LLS	DM-NLS	DM-LLS
(a) Least reflection	-178.57%	-251.26%	-100.30%	-152.57%
(b) Corridor reflection	55.03%	50.69%	37.76%	31.76%
(c) Corner reflection	63.64%	59.08%	48.34%	41.86%
(d) Surround reflection	69.23%	68.35%	58.39%	57.20%
(e) Heterogeneous reflection	67.68%	56.32%	57.36%	42.36%
(f) Least reflection (tilting PD)	69.59%	62.71%	61.23%	52.46%
(g) Surround reflection (tilting PD)	60.38%	59.62%	48.28%	47.29%
(h) Heterogeneous reflection (tilting PD)	63.54%	57.38%	52.53%	44.51%

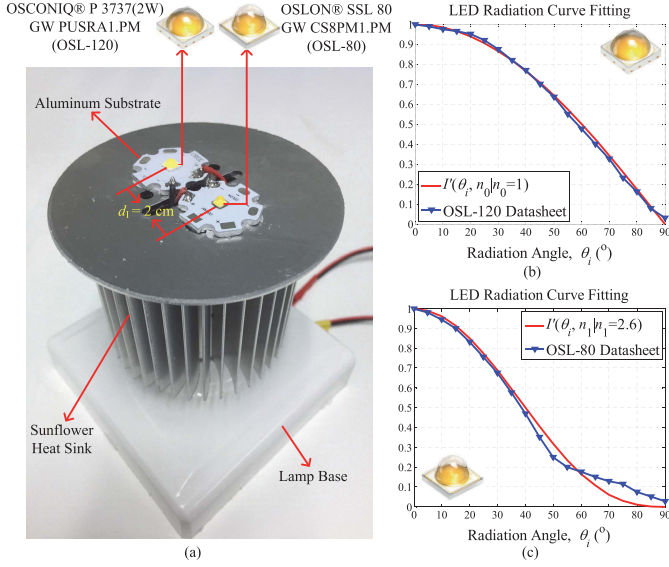


Fig. 8. The implemented prototype DM-LED lamp using two OSRAM® LED chips [40], [41] with different RLMNs.

frequencies due to spectral leakages, a sampling frequency of 96 kHz was set for the laptop's sound card.

To quantify the positioning performance of the prototype system, we divided the square region projected by the four DM-LED lamps into a total of 9×9 grids, each of which took a size of $15 \text{ cm} \times 15 \text{ cm}$. Then we obtained eight spectral amplitudes, which corresponded to the eight LED chips, in one processing time window after sampling at each grid. By traversing all 81 grids and processing a number of 81×8 spectral amplitudes, the LE distribution of all grids was obtained. Furthermore, we should minimize the difference between the output powers of the eight LEDs of the four DM-LED lamps, by adjusting the driving currents from the direct current power supplies for the eight LED channels. For fair comparison, the power of conventional lamp is approximately set to be equal to the summed power of both LEDs of DM-LED lamp.

3) *Performance Measurements*: As shown in Fig. 10, we set up a few realistic scenarios to measure the LE performances of RSS-nVLP and DM-nVLP systems. Specifically, four glossy white boards were first exploited to imitate three typical reflection scenarios, namely the corridor reflection, the corner reflection and the surround reflection environments. For comparison

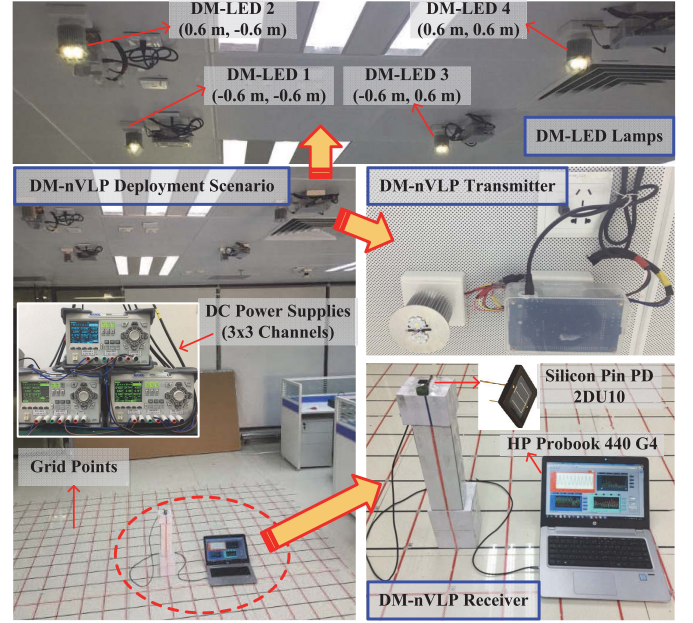


Fig. 9. The experimental deployment scenario of the prototype DM-nVLP system.

purpose, the least reflection case set up in an open area dispensing with any white boards was also tested. Moreover, a metal shelf with a size of $1.5 \text{ m} \times 0.4 \text{ m} \times 1.75 \text{ m}$, an oak texture board, a grained white texture board and a mahogany texture board were utilized to construct a heterogeneous reflection scenario, where a person stood at a random position in the testing area without blocking the LOS links from the DM-LED lamp to the PD. The size of each board was $1.2 \text{ m} \times 2.4 \text{ m}$. The LE performances of the RSS-nVLP and DM-nVLP systems measured in the eight scenarios of Fig. 10 are illustrated in Fig. 11.

A comparison of Fig. 7 and Fig. 11 shows that a typical difference of about 1-4 cm exists between the simulation and the experimental results, due to the inevitable practical impairments such as the non-ideal d_t , the imperfect uniformity of LEDs' luminance, and measurement errors. Specifically, consistent with the corresponding simulated case in Fig. 7(a), Fig. 11(a) indicates that RSS-nVLP systems provide a better measured LE performance than DM-nVLP systems in the scenario with few reflections. In contrast, Fig. 11(b), Fig. 11(c), Fig. 11(d) and Fig. 11(e) reveal that in the scenarios with more



Fig. 10. The experimental setups of various reflection scenarios.

reflections, both DM-NLS and DM-LLS systems substantially outperform their RSS counterparts, reducing the average measured LE by about 30-70% across all grids, as summarized in Table IV. Again, similar to Fig. 7, we can see from Fig. 11 that DM-nVLP systems perform much more stably than RSS-nVLP systems do, in all but least reflection scenarios. Particularly, the performances of RSS-nVLP systems significantly deteriorate due to heavier multi-path reflections at the locations closer to the boards. Among the two proposed schemes, DM-NLS typically outperforms DM-LLS at the cost of an increased complexity.

Furthermore, since the RSSR defined by (13), α_i , is independent of the incident angle φ_i between the i -th DM-LED lamp and the PD, it implies that the system's performance would not be affected by a PD tilts under a given FOV constraint. This is verified by Fig. 11(f), Fig. 11(g) and Fig. 11(h), where we tested the various schemes with the PD tilting 16.5° along a random orientation. For example, comparing Fig. 11(f) with Fig. 11(a), we note that the RSS-nVLP systems, which require that the normals of the LED's and the PD's surfaces are parallel, suffer a large performance degradation when the PD tilts. However, the performances of DM-nVLP systems remain statistically stable regardless of the PD tilt, as seen in Table IV, thus beneficially relaxing the constraint on the UE's posture in practical applications. For instance, when a person wearing or holding an nVLP UE walks in the room, the UE and its PD may randomly tilt. In this case, the proposed DM-nVLP scheme would perform much better than its conventional RSS-nVLP counterpart. Moreover, note that there exhibit some high error peaks at certain locations, printed in red color, in the two RSS-LLS sub-figures of Fig. 11(h). The reason for this phenomenon is that in such a heterogeneous reflection scenario, the tilting position of the PD and the presence of the pedestrian may together largely impair the achievable RSS at

specific part of the area, thus dramatically degrading the LE performance of the RSS-nVLP system. In the same scenario, however, the simultaneous reduction of the RSSs from both LEDs of a DM-LED lamp may not significantly affect the ratio between them, or more precisely the RSSR defined by (13) in the proposed DM-nVLP scheme. Therefore, there exists no distinct error peaks for the DM-nVLP system in the hostile heterogeneous reflection scenario, as demonstrated by the two DM-nVLP sub-figures of Fig. 11(h).

Finally, we summarize the benefits of the proposed DM-nVLP solution compared to existing schemes:

- By approximating the multi-path reflection related parameters as a constant, the DM-nVLP system can robustly mitigate the adverse impact from NLOS propagations that cannot be handled by RSS-nVLP. This is beneficial when enabling VLP services in for example supermarkets, where the multi-path reflections can be quite severe in the complicated shopping environment, especially with a high density of reflective shelves and goods.
- Based on (8) and (13), the proposed RSSR-based mechanism eliminates the need of a few receiver-dependent *a priori* parameters that are difficult to obtain in practical scenarios, for instance ℓ , β and φ_i , thus offering an increased flexibility in practical applications. Furthermore, a moderately tilting PD may have little impact on the system's performance, which is a desirable merit for mobile applications, where for example a wearable UE is used by a pedestrian.
- Furthermore, when deployed in an environment with any level of reflections, our scheme operates online and avoids the excessive, thus practically unrealistic offline data collection campaign for fingerprinting all possible reflection strengths in the case of conventional RSS-nVLP systems.

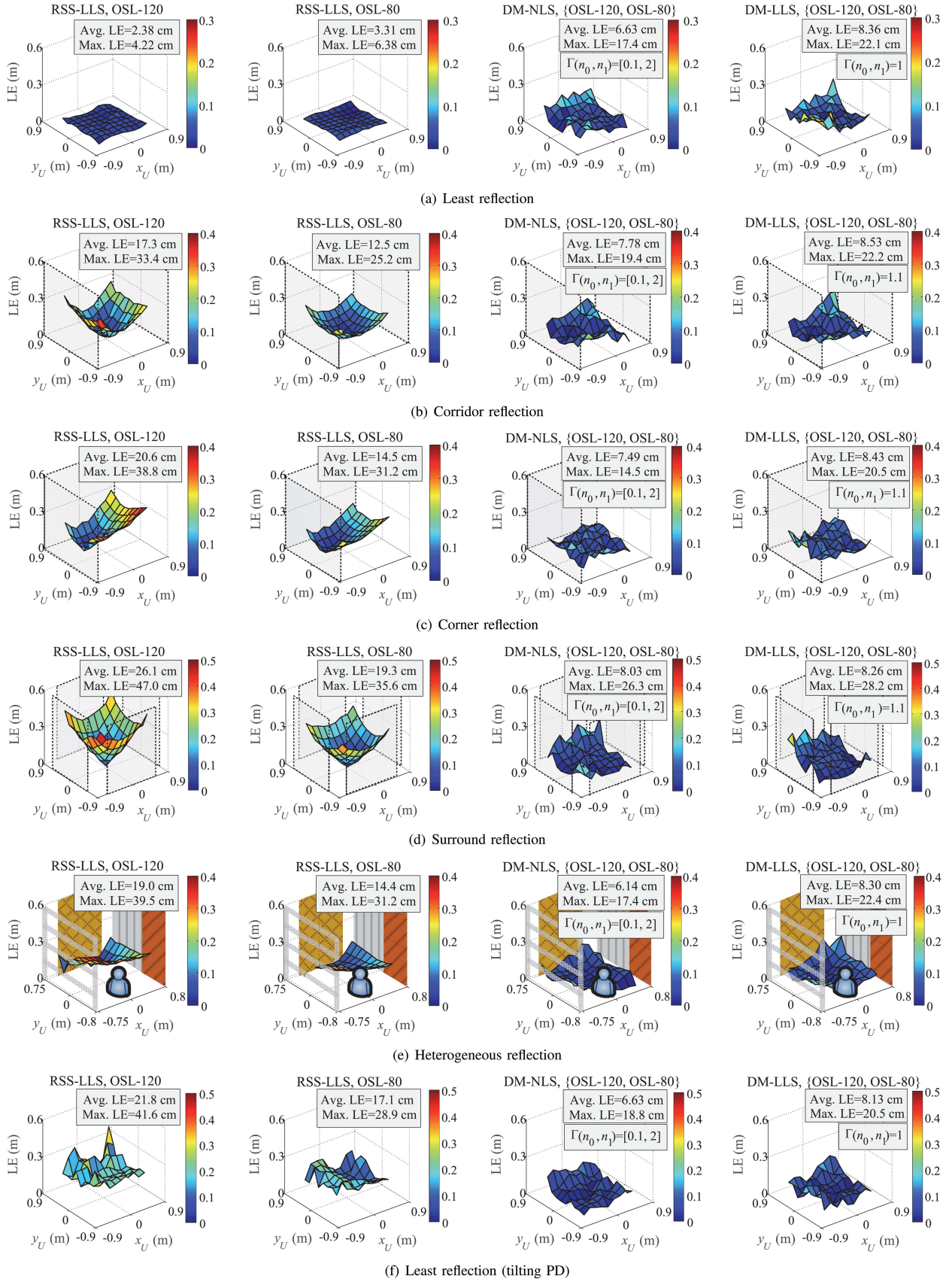


Fig. 11. The measured LE performances of RSS-nVLP and DM-nVLP systems in various environments, where we set $\{I_L, h\} = \{1.2 \text{ m}, 1.83 \text{ m}\}$, and a PD tilt of 16.5° along random orientations in (f), (g) and (h). Note that the pedestrian icon in (e) and (h) only indicates the presence of a pedestrian and does not represent his specific location.

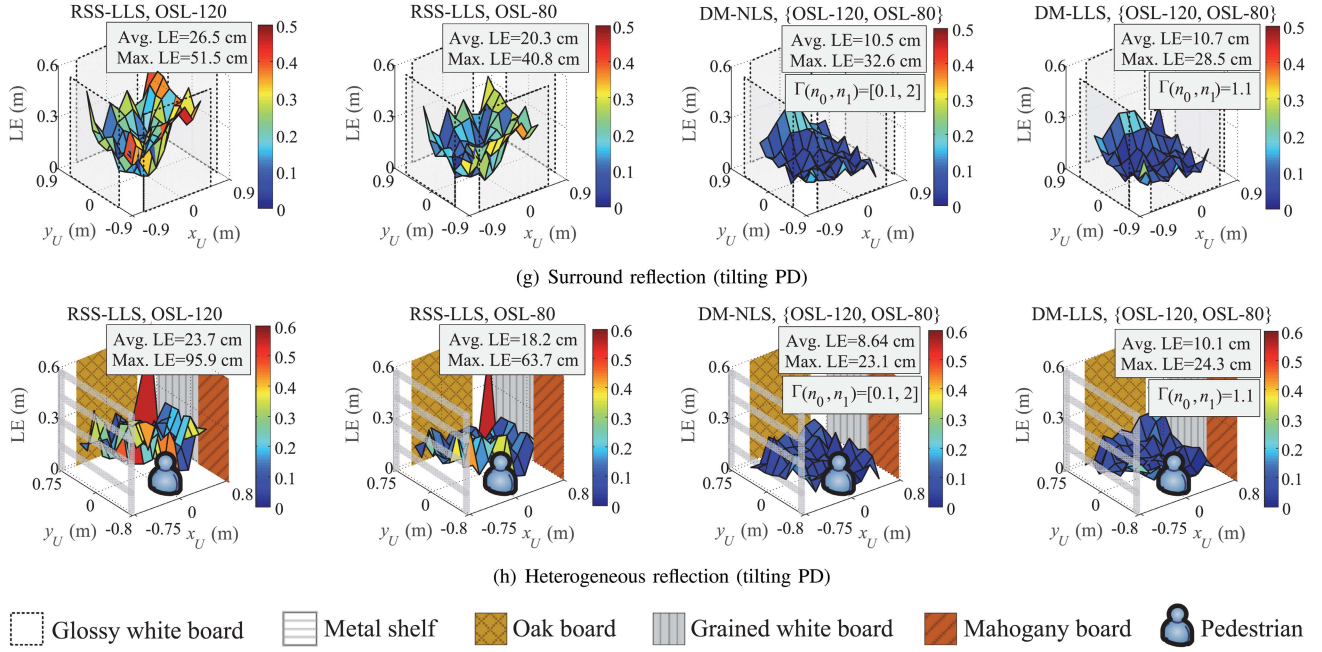


Fig. 11. (Continued) The measured LE performances of RSS-nVLP and DM-nVLP systems in various environments, where we set $\{I_L, h\} = \{1.2 \text{ m}, 1.83 \text{ m}\}$, and a PD tilt of 16.5° along random orientations in (f), (g) and (h). Note that the pedestrian icon in (e) and (h) only indicates the presence of a pedestrian and does not represent his specific location.

VI. CONCLUSION

In this paper, a novel DM-nVLP model considering multi-path propagations is proposed for indoor positioning. The scheme exploits LEDs of different RLMNs, a new RSSR-aided radiation angle measurement mechanism and an artificial parameter, namely the channel gain ratio factor, which together help to decouple the correlation between the UE's location and the unknown NLOS channel gains, thus facilitating efficient operations of the DM-NLS/LLS algorithms. Theoretical analysis on the CRLB, extensive simulation and experimental results are provided. It is shown that the DM-nVLP system can provide a significantly improved positioning performance over the conventional RSS-nVLP scheme, exhibiting increased robustness and flexibility, and thus a great potential for practical applications especially under scenarios with multi-path reflections.

APPENDIX A

DERIVATION OF $\Gamma(S_i, U, E_N, n_0, n_1)$ OF (19)

In this Appendix, we aim to derive the expression of $\Gamma(S_i, U, E_N, n_0, n_1)$ based on the iterative site-based model [31], [32]. Let $\theta_{S_i, \varepsilon_p^{(1)}}$, $\varphi_{S_i, \varepsilon_p^{(1)}}$ and $d_{S_i, \varepsilon_p^{(1)}}$ be the radiation angle, the incident angle and the distance between the i -th DM-LED source S_i and the micro-reflective area $\varepsilon_p^{(1)}$, respectively, where $p = 1, \dots, N$. For notation simplicity, we again omit E_N and n_τ in most variables.

According to Fig. 2, based on (1), (6) and (17), we have

$$\begin{aligned} H_{S_i, U}^{(1)} &= \sum_{p=1}^N \varrho_{\varepsilon_p^{(1)}} H_{S_i, \varepsilon_p^{(1)}}^{(0)} H_{\varepsilon_p^{(1)}, U}^{(0)} \end{aligned}$$

$$\begin{aligned} &= \sum_{p=1}^N \varrho_{\varepsilon_p^{(1)}} \cdot \frac{I(\theta_{S_i, \varepsilon_p^{(1)}}, n_\tau) g_\varepsilon(\varphi_{S_i, \varepsilon_p^{(1)}})}{P^2 d_{S_i, \varepsilon_p^{(1)}}^2} \cdot H_{\varepsilon_p^{(1)}, U}^{(0)} \\ &= \frac{n_\tau + 1}{2\pi P^2} \sum_{p=1}^N \cos^{n_\tau} \theta_{S_i, \varepsilon_p^{(1)}} \cdot \left[\frac{\varrho_{\varepsilon_p^{(1)}} g_\varepsilon(\varphi_{S_i, \varepsilon_p^{(1)}})}{d_{S_i, \varepsilon_p^{(1)}}^2} H_{\varepsilon_p^{(1)}, U}^{(0)} \right] \\ &= \frac{n_\tau + 1}{2\pi P^2} \sum_{p=1}^N \cos^{n_\tau} \theta_{S_i, \varepsilon_p^{(1)}} \cdot \chi_{p, S_i, U}^{(1)} \end{aligned} \quad (48)$$

and similarly

$$\begin{aligned} H_{S_i, U}^{(2)} &= \sum_{p=1}^N \varrho_{\varepsilon_p^{(1)}} \cdot H_{S_i, \varepsilon_p^{(1)}}^{(0)} \cdot H_{\varepsilon_p^{(1)}, U}^{(1)} \\ &= \frac{n_\tau + 1}{2\pi P^2} \sum_{p=1}^N \cos^{n_\tau} \theta_{S_i, \varepsilon_p^{(1)}} \cdot \chi_{p, S_i, U}^{(2)}, \end{aligned} \quad (49)$$

where $\varrho_{\varepsilon_p^{(k)}}$ denotes the reflection coefficient of $\varepsilon_p^{(k)}$, $g_\varepsilon(\cdot)$ represents the optical gain function of the micro-reflective area [31], and we have

$$\chi_{p, S_i, U}^{(1)} = \frac{\varrho_{\varepsilon_p^{(1)}} g_\varepsilon(\varphi_{S_i, \varepsilon_p^{(1)}})}{d_{S_i, \varepsilon_p^{(1)}}^2} H_{\varepsilon_p^{(1)}, U}^{(0)} \quad (50)$$

and

$$\begin{aligned} \chi_{p, S_i, U}^{(2)} &= \frac{\varrho_{\varepsilon_p^{(1)}} g_\varepsilon(\varphi_{S_i, \varepsilon_p^{(1)}})}{d_{S_i, \varepsilon_p^{(1)}}^2} H_{\varepsilon_p^{(1)}, U}^{(1)} \\ &= \frac{\varrho_{\varepsilon_p^{(1)}} g_\varepsilon(\varphi_{S_i, \varepsilon_p^{(1)}})}{d_{S_i, \varepsilon_p^{(1)}}^2} \sum_{p=1}^N \varrho_{\varepsilon_p^{(2)}} H_{\varepsilon_p^{(1)}, \varepsilon_p^{(2)}}^{(0)} H_{\varepsilon_p^{(2)}, U}^{(0)}. \end{aligned} \quad (51)$$

Based on the inductive method with the aid of (48) and (49), the general expression of $H^{(k)}(S_i, U, E_N, n_\tau)$ may

be formulated as

$$H_{S_i,U}^{(k)} = \frac{n_\tau + 1}{2\pi P^2} \sum_{p=1}^N \cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}} \cdot \chi_{p,S_i,U}^{(k)}, \quad (52)$$

where

$$\chi_{p,S_i,U}^{(k)} = \frac{\varrho_{\varepsilon_p^{(1)}} g_\varepsilon(\varphi_{S_i,\varepsilon_p^{(1)}})}{d_{S_i,\varepsilon_p^{(1)}}^2} H_{\varepsilon_p^{(1)},U}^{(k-1)} \quad (53)$$

is a term that does not depend on the RLMN n_τ . For notation brevity, $\chi_{p,S_i,U}^{(k)}$ is abbreviated as $\chi_p^{(k)}$ in the sequel.

Thus, by utilizing (3), (5), (6), (7) and (52), we arrive at

$$\begin{aligned} \rho(S_i, U, E_N, n_\tau) &= \frac{H_{\text{NLOS}}(S_i, U, E_N, n_\tau)}{H_{\text{LOS}}(S_i, U, n_\tau)} \\ &\approx \frac{H_{S_i,U}^{(1)} + H_{S_i,U}^{(2)} + \dots + H_{S_i,U}^{(K)}}{H_{S_i,U}^{(0)}} \\ &= \frac{\frac{n_\tau+1}{2\pi P^2} \sum_{p=1}^N [\cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}} \cdot (\chi_p^{(1)} + \chi_p^{(2)} + \dots + \chi_p^{(K)})]}{\frac{n_\tau+1}{2\pi d_i^2} A_U \cos^{n_\tau} \theta_i \cdot \cos^m \varphi_i} \\ &= \frac{h^2 \sum_{p=1}^N [\cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}} \cdot (\chi_p^{(1)} + \chi_p^{(2)} + \dots + \chi_p^{(K)})]}{A_U P^2 \cos^{(n_\tau+m+2)} \theta_i}, \end{aligned} \quad (54)$$

where $\theta_{S_i,\varepsilon_p^{(1)}}, \chi_p^{(k)}$ ($p = 1, \dots, N$) are two random variables, which are independent due to the random spatial distribution of the NLOS propagation paths associated with the numerous micro-reflective areas inside the room. Assuming that the light emission from a DM-LED lamp is uniform in the space, $\theta_{S_i,\varepsilon_p^{(1)}}$ may be considered as satisfying the uniform distribution within $[0, \frac{\pi}{2})$, with a PDF given by

$$f(\theta_{S_i,\varepsilon_p^{(1)}}) = \begin{cases} \frac{2}{\pi}, & 0 \leq \theta_{S_i,\varepsilon_p^{(1)}} < \frac{\pi}{2} \\ 0, & \text{otherwise.} \end{cases} \quad (55)$$

Hence, we can obtain

$$\begin{aligned} E[\cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}}] &= \int_{-\infty}^{+\infty} \cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}} f(\theta_{S_i,\varepsilon_p^{(1)}}) d\theta_{S_i,\varepsilon_p^{(1)}} \\ &= \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}} d\theta_{S_i,\varepsilon_p^{(1)}}, \end{aligned} \quad (56)$$

where $E[\cdot]$ is the expectation operator. Since N typically takes a very large value, we have

$$\begin{aligned} \sum_{p=1}^N \cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}} \cdot (\chi_p^{(1)} + \chi_p^{(2)} + \dots + \chi_p^{(K)}) \\ \approx N \cdot E[\cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}} \cdot (\chi_p^{(1)} + \chi_p^{(2)} + \dots + \chi_p^{(K)})]. \end{aligned} \quad (57)$$

By exploiting (56) and (57), we may further develop (54) as (58), shown at the bottom of the page, where $\bar{\chi}^{(k)} = E[\chi_p^{(k)}]$, $p = 1, \dots, N$, $k = 1, \dots, K$, and $\chi_c = \frac{2h^2 N(\bar{\chi}^{(1)} + \bar{\chi}^{(2)} + \dots + \bar{\chi}^{(K)})}{\pi A_U P^2}$. With the aid of (53), we note that $\chi_c > 0$. Then, by substituting (58) into (14), we have

$$\begin{aligned} \Gamma(S_i, U, E_N, n_0, n_1) &= \frac{1 + \rho(S_i, U, E_N, n_0)}{1 + \rho(S_i, U, E_N, n_1)} \approx \frac{\frac{1}{\chi_c} + \frac{\int_0^{\frac{\pi}{2}} \cos^{n_0}(\theta_{S_i,\varepsilon_p^{(1)}}) d\theta_{S_i,\varepsilon_p^{(1)}}}{\cos^{(n_0+m+2)} \theta_i}}{\frac{1}{\chi_c} + \frac{\int_0^{\frac{\pi}{2}} \cos^{n_1}(\theta_{S_i,\varepsilon_p^{(1)}}) d\theta_{S_i,\varepsilon_p^{(1)}}}{\cos^{(n_1+m+2)} \theta_i}}. \end{aligned} \quad (59)$$

In order to further simplify (59), we exploit [38, (98)]

$$\int_0^{\frac{\pi}{2}} (\sin \theta)^{2x-1} (\cos \theta)^{2y-1} d\theta = \frac{\gamma(x) \gamma(y)}{2\gamma(x+y)}, \quad (60)$$

where $\gamma(x) = \int_0^{+\infty} \nu^{x-1} e^{-\nu} d\nu$ is the Gamma function [38]. Let $2x - 1 = 0$ and $2y - 1 = n_\tau$ ($\tau = \{0, 1\}$), using variable substitution, we can transform (60) to

$$\int_0^{\frac{\pi}{2}} \cos^{n_\tau} \theta d\theta = \frac{\gamma(\frac{1}{2}) \gamma(\frac{n_\tau+1}{2})}{2\gamma(\frac{n_\tau+1}{2})} = \frac{\sqrt{\pi} \gamma(\frac{n_\tau+1}{2})}{2\gamma(\frac{n_\tau}{2} + 1)}. \quad (61)$$

Finally, after substituting θ with $\theta_{S_i,\varepsilon_p^{(1)}}$ in (61) and then inserting (61) into (59), we arrive at (19).

$$\begin{aligned} \rho(S_i, U, E_N, n_\tau) &= \frac{h^2 N \cdot E[\cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}} \cdot (\chi_p^{(1)} + \chi_p^{(2)} + \dots + \chi_p^{(K)})]}{A_U P^2 \cos^{(n_\tau+m+2)} \theta_i} \\ &= \frac{h^2 N \cdot E[\cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}}] \cdot \{E[\chi_p^{(1)}] + E[\chi_p^{(2)}] + \dots + E[\chi_p^{(K)}]\}}{A_U P^2 \cos^{(n_\tau+m+2)} \theta_i} \\ &= \frac{2h^2 N(\bar{\chi}^{(1)} + \bar{\chi}^{(2)} + \dots + \bar{\chi}^{(K)})}{\pi A_U P^2} \cdot \frac{\int_0^{\frac{\pi}{2}} \cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}} d\theta_{S_i,\varepsilon_p^{(1)}}}{\cos^{(n_\tau+m+2)} \theta_i} \\ &= \chi_c \cdot \frac{\int_0^{\frac{\pi}{2}} \cos^{n_\tau} \theta_{S_i,\varepsilon_p^{(1)}} d\theta_{S_i,\varepsilon_p^{(1)}}}{\cos^{(n_\tau+m+2)} \theta_i}, \end{aligned} \quad (58)$$

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